Space–Time Stokes–Dirac Discretization

Affine Pairing Gauges, Constitutive Cayley Coordinates,
and Active Interior Reduction

Redha Moulla
AXIA, France
redha.moulla@axia-conseil.com

August 2026

Abstract

We classify compatible pairings on a finite-dimensional discrete space–time cylinder. For a possibly nonsurjective differential $D : E \to F$, compatibility exists precisely when the boundary form annihilates $\ker D$, and the active freedom is $\Lambda^2(E/\ker D)^*$. On a temporal–spatial complex this separates temporal, spatial, and intrinsic mixed gauges. A regular interior solve then yields an exact sequence for their visible and invisible cap reductions.

For nonsingular quadratic Hamiltonian systems, cap gauges are shifted Cayley coordinates on the no-$+1$ part of the energy-orthogonal group; we determine their exact symplectic locus and defect. An active polynomial cell $\mathcal{P}_s \to \mathcal{P}_{s-1}$ generates the Gauss–s cap map by elimination. On the 2012 polynomial transmission line, mixed cylinder lifts preserve cap–mantle power, dissipation, and interface cancellation. Every alternating cap gauge also admits a minimal auxiliary-mode Schur generation whose interior dimension equals its rank; this algebraic realization is distinguished from selection by an independently prescribed field cell. For each fixed semidiscrete generator, a unique central coefficient raises operator convergence from $2s$ to $2s + 2$; a recursive rational hierarchy gives order at least $2s + 2r$, exact when the next coefficient is nonzero, while noncentral additions retain the attained order and leave the scalar functional calculus. Its reduced polynomial form separates removable Cayley-chart resonances from true poles and shows unit amplification on oscillatory modes but not general A-stability. The exact coordinate is a Stieltjes function in z^2, and C_s is its diagonal or near-diagonal Padé approximant. These Gauss coordinates have positive Jacobi-chain realizations, whereas the corrected hierarchy need not: the first nontrivial member already violates Stieltjes positivity. Executable tests verify the reductions, power laws, asymptotic coefficients, high-order branches, and a multiprecision noncommutation law.

Keywords. Port-Hamiltonian systems; Stokes–Dirac structures; space–time complexes; discrete gradients; Gauss collocation; symplecticity; power balance; geometric numerical integration.

MSC 2020. 37M15; 65P10; 65M60; 93C20.

1 Introduction

Distributed port-Hamiltonian systems expose boundary power as part of the geometry rather than as an afterthought [33, 17]. On a space–time cylinder, the initial and final energies and the lateral power ports are different components of one oriented boundary. A discretization that starts from this cylinder should therefore not be assembled by merely attaching a time index to a spatial Stokes matrix. It should possess a total discrete differential, a global boundary form, and a compatible pairing before a constitutive law is selected.

Existing structure-preserving spatial discretizations construct a finite-dimensional Stokes identity whose boundary term becomes a power port [20, 29, 4]. In particular, discrete-exterior-calculus constructions produce simplicial Dirac structures on space while retaining continuous time; treating space and time on equal footing is a distinct extension already identified in that literature [29]. Passivity-consistent time discretizations and energy-consistent Petrov–Galerkin formulations provide complementary temporal closures [5, 12]. Symplectic Runge–Kutta theory and collocation identify Gauss–Legendre methods as the canonical collocation schemes preserving quadratic invariants [8, 28, 27, 14]. Discrete-gradient methods instead preserve a prescribed Hamiltonian exactly, including for nonlinear Hamiltonians [21, 13, 26, 10]. High-order segment and line-integral constructions connect this viewpoint to collocation and Gaussian quadrature [15, 3, 25, 16]. Locally exact discrete-gradient modifications likewise use step- and linearization-dependent skew operators to reproduce a target linearized flow while preserving energy [9]. The overlap is the use of a constitutively adapted skew operator. The present hierarchy is not claimed as the first such modification: its additional content is the affine Stokes classification before closure, the exact volume-to-cap sequence, the central/noncentral split, and the realization of either direction by a mixed cylinder two-form.

There are established genuinely space–time routes with different primary structures. Covariant variational integrators derive a discrete multisymplectic form formula from a discrete action [23]; space–time finite elements provide arbitrary-order energy and momentum laws for multisymplectic PDEs [6]; and HDG/FEEC formulations characterize local multisymplecticity on numerical traces [22, 31, 32]. The present construction is not a claim of priority over those field discretizations and does not assume a discrete action. Its different question is to classify all pairings compatible with a prescribed discrete Stokes formula, transport their affine freedom through a cell reduction, and identify which mixed gauges generate boundary maps outside the scalar method-of-lines class.

The common algebraic fact is simple but consequential: a Stokes identity or a discrete chain rule fixes only the symmetric part of a pairing. The remaining freedom is alternating on the quotient by directions invisible to the differential. The central question is whether this observation can be made directly on the space–time cylinder, so that the spatial freedom found in [20], the temporal discrete-gradient gauge, and mixed nonseparable pairings become sectors of one object.

This paper answers that question at eight levels.

1. For an arbitrary, possibly nonsurjective, differential $D : E \to F$, the active compatible pairings form an affine space over $\Lambda^2(E/\ker D)^*$. Applied to the genuine total complex through a degree-respecting symmetric double, this gives a bilinear gauge between complementary active quotients. Applied instead to the stacked one-bidegree cylinder map, it gives temporal, spatial, and mixed sectors.

2. Once a Dirac relation, constitutive closure, boundary causality, and regular interior solve have fixed a reconstruction $L_h : Q_\partial \to Q_{\mathrm{st}}$, the exact volume-to-cap reduction is $\omega \mapsto L_h^*\omega$. We classify its image and kernel and derive an intrinsic rank criterion for the visibility of mixed gauges. For trace-generated forms, a raw weak cell containing the same volume matrix is assembled and exactly realizes the prescribed cap gauge. We state explicitly that this is a realization theorem, not a selection of the gauge by independent interior dynamics.

3. For a single time step and a quadratic Hamiltonian, the family $\Lambda^2 V^*$ gives constitutively shifted Cayley coordinates on the no-$+1$ cell of $O(Q)$. Exact power holds on the whole chart; its symplectic intersection and factorized defect are computed explicitly, and the exact-flow gauge is obtained in closed form.

4. For an s-stage Runge–Kutta reconstruction with invertible stage matrix, all compatible

bilinear stage pairings are classified. Local diagonal pairing reduces compatibility to the Cooper condition, and a short spanning argument proves directly that symplectic collocation is precisely Gauss collocation. Conversely, starting from the polynomial temporal cell $\mathcal{P}_s \to \mathcal{P}_{s-1}$, we retain the Gauss stages as active interior variables and prove that their elimination produces the diagonal Padé cap map and its induced two-point gauge.

5. For a space–time Stokes–Dirac discretization, Dirac maximality is kept separate from the constitutive chain rule. This yields exact cap energy, lateral boundary power, passivity with dissipation, and exact cancellation on glued faces. Under scalar locality, the stacked tensor equations force factorization, and maximal exactness then selects Gauss–Legendre on each axis.

6. A general rectangular Stokes theorem shows that an invertible effort-and-input reconstruction produces a square open representation with skew interconnection, dual input/output blocks, and skew feedthrough. This turns the spatial reduction used in the 2012 line from a numerical observation into a direct algebraic consequence.

7. On that rectangular transmission-line reduction, a two-form on spatial derivatives of temporal increments gives an explicit mixed cylinder gauge. For every s, its cap reduction gives a reversible, energy-preserving deformation of Gauss–s. In the fixed one-term ansatz, one central coefficient and only one raises the order from $2s$ to exactly $2s+2$; every other fixed coefficient has exactly order $2s$. Adding the next odd-power gauge term preserves that higher order while permitting noncommutation, and a recursive central hierarchy yields successive two-order lower bounds, exact under an explicit non-annulation condition. A commutator criterion proves when the resulting method cannot belong to the scalar functional calculus of the semidiscrete line. A dispersion experiment turns the gauge into a design variable.

8. Every finite-rank alternating cap gauge has a minimal auxiliary Schur generator with interior dimension equal to its rank. For the central scalar coordinates, Stieltjes–Padé theory gives a stronger positive Jacobi-chain realization of every Gauss member. The corrected $(s, r) = (2, 1)$ coordinate fails this positivity test, separating unrestricted algebraic generation from passive interior generation.

The last level exposes the following structural chain without identifying its distinct notions of positivity:

$$\boxed{\begin{array}{c}\text{Gauss} \iff \text{Padé–Stieltjes} \iff \text{finite positive measure} \\ \iff \text{positive Jacobi resolvent} \iff \text{Stieltjes continued fraction.}\end{array}} \tag{1}$$

Scope and novelty. The affine Stokes principle, classical Gauss quadrature, symplectic Runge–Kutta theory, discrete gradients, and the Dirac construction are established ingredients. The new contribution is their common quotient-space formulation on a discrete cylinder: the nonsurjective classification, the explicit distinction between the total and stacked differentials, the exact visibility sequence for volume-to-cap reduction, the intrinsic mixed reduction, the constitutively shifted Cayley coordinates of cap gauges, the rectangular reconstruction theorem, the active-stage derivation of the Gauss cap gauge, the sharp auxiliary Schur existence-and-minimality construction, the positive Jacobi realization and its obstruction, the symplectic-locus and defect formulas, the exact one-term order dichotomy, its multi-term correction hierarchy, tensor rigidity derived from the global Stokes equations, and the mixed Gauss–s family with its scalar-functional-calculus obstruction. The trace-generated theorem is deliberately described as exact realization; only the higher-degree stage cell derives its cap gauge from an independent

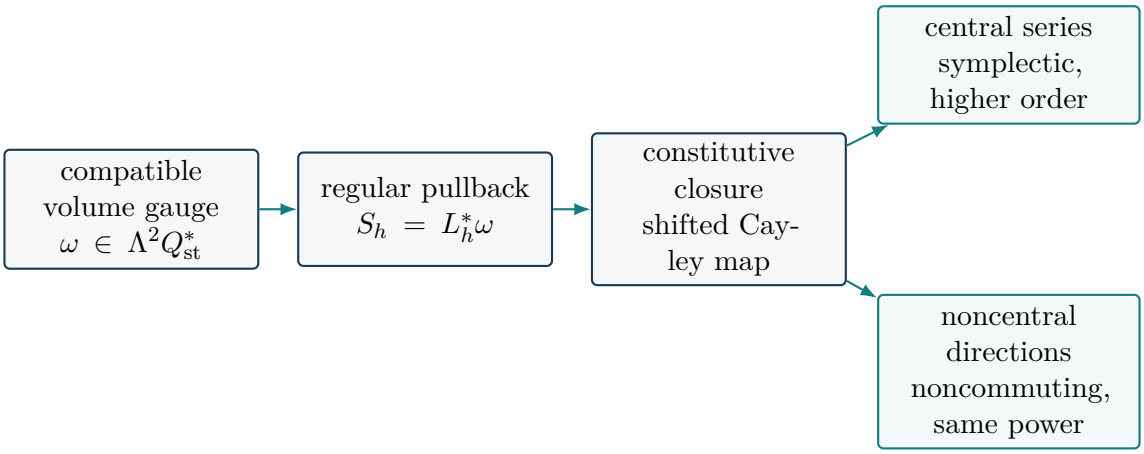


Figure 1: Architecture of the construction. Compatibility is classified in the volume, transported to the caps, and closed through a Cayley chart. Only after the constitutive generator $A_N = \mathcal{J}_N Q_N$ is fixed does the cap space split operationally into a central accuracy hierarchy and noncentral design directions; both retain the same contracted power identity.

interior solve. We do not call an arbitrary compatible bilinear form a multi-Dirac structure. Multi-Dirac field theory includes additional graded geometric and integrability requirements [34].

Order convention. Every order statement below concerns matrix/operator convergence to e^{hA_N} as $h \to 0$ for a *fixed finite-dimensional linear semidiscrete generator* A_N, with constants allowed to depend on N. It is neither a universal Runge–Kutta order statement for nonlinear vector fields nor a simultaneous space–time convergence theorem. The nonlinear section establishes the chain rule and energy identity, not the linear hierarchy's order.

Definition 1.1 (Local matrix defect and global operator order)**.** *For fixed A_N, a one-step family M_h has* local matrix defect of order $p+1$ *if*

$$\left\| M_h - e^{hA_N} \right\| = O(h^{p+1}) \qquad (h \to 0) \tag{2}$$

in any matrix norm. It has global operator order p *on bounded time intervals if, for each fixed $T > 0$ and $n = T/h \in \mathbb{N}$,*

$$\left\| M_h^n - e^{TA_N} \right\| = O(h^p). \tag{3}$$

For a consistent finite-dimensional family satisfying Equation (2), the telescoping identity

$$M_h^n - e^{TA_N} = \sum_{k=0}^{n-1} M_h^{n-1-k} (M_h - e^{hA_N}) e^{khA_N}$$

gives Equation (3). Thus the one-power shift used below is an explicit fixed-A_N matrix estimate.

Terminology for "exact." The adjective is always qualified by its object. *Exact power* or *exact energy* means an algebraic identity at the discrete level; *exact elimination* means equality of two finite-dimensional Schur reductions; *exact order* means that the displayed leading error coefficient is nonzero; and *exact-flow gauge* refers specifically to the Cayley coordinate of e^{hA_N}. None of these phrases asserts a simultaneous continuum limit unless such a limit is stated separately.

Notation. All spaces are finite-dimensional and real. A vector space V and its dual V^* are distinguished until coordinates are chosen. In coordinates, covectors are represented by columns and paired by transpose. For an invertible skew matrix J, we write $\Omega = J^{-1}$. The state dimension is n; in symplectic statements $n = 2m$.

2 The affine Stokes-pairing principle

We first isolate the algebra that is shared by spatial integration by parts and the temporal discrete chain rule.

Definition 2.1 (Discrete Stokes data)**.** *Let E, F be vector spaces, let $D : E \to F$ be linear, and let $b : E \times E \to \mathbb{R}$ be a symmetric bilinear boundary form. A bilinear map $M : E \times F \to \mathbb{R}$ is* Stokes compatible *if*

$$M(z, Dw) + M(w, Dz) = b(z, w) \qquad (z, w \in E). \tag{4}$$

Theorem 2.2 (Active affine Stokes family)**.** *Let $K = \ker D$, $I = \operatorname{im} D$, and $r = \operatorname{rank} D$. A compatible pairing exists if and only if*

$$b(k, \ell) = 0 \qquad (k, \ell \in K). \tag{5}$$

On the active flow space I, if M_0 is one compatible pairing, all compatible restrictions are

$$M_\omega(z, Dw) = M_0(z, Dw) + \omega([z], [w]), \qquad \omega \in \Lambda^2(E/K)^*. \tag{6}$$

Consequently the active solution set is an affine space of dimension

$$\frac{r(r-1)}{2}. \tag{7}$$

If pairings are required on all of $E \times F$, their values transverse to I are unconstrained. If $\mathcal{H}$ denotes the full homogeneous direction, it fits into the short exact sequence

$$0 \longrightarrow \mathrm{Bil}(E, F/I; \mathbb{R}) \longrightarrow \mathcal{H} \longrightarrow \Lambda^2(E/K)^* \longrightarrow 0. \tag{8}$$

Every choice of a complement of I in F splits this sequence, but the splitting is not canonical. The full direction has dimension $r(r-1)/2 + \dim(E)(\dim(F) - r)$.

Proof. Necessity follows from Equation (4) with $z, w \in K$. Assume Equation (5). Choose a section $R : I \to E$ of $D : E \to I$, and write $z = k_z + RDz$. Define, for $g \in I$,

$$M_0(z, g) = \tfrac{1}{2} b(RDz, Rg) + b(k_z, Rg).$$

Substitution and Equation (5) verify compatibility on I. Choosing a complement of I in F and extending arbitrarily produces a pairing on all of $E \times F$.

If M and M_0 are compatible, their difference $N = M - M_0$ obeys

$$N(z, Dw) + N(w, Dz) = 0.$$

Taking $w \in K$ gives $N(w, Dz) = 0$ for every z, hence N factors through E/K in its first argument. The induced bijection $\bar{D} : E/K \to I$ identifies the two active arguments, so the induced bilinear form is alternating. Conversely, every alternating form in Equation (6) cancels from the symmetric identity. This proves the active classification. Finally, compatibility evaluates the second argument only on I. Restriction to I maps $\mathcal{H}$ onto $\Lambda^2(E/K)^*$, and its kernel consists precisely of bilinear forms vanishing on I, equivalently forms on $E \times (F/I)$. This proves the exact sequence and the dimension formula. □

Remark 2.3. *The Stokes-visible active gauge is intrinsically an alternating two-form on the quotient $E/\ker D$, not a skew matrix until a basis and an identification with its dual are selected. The kernel term in Equation (8) is invisible to Stokes compatibility and should not be counted as an active gauge. This distinction becomes essential for a space–time differential, whose image is constrained by integrability.*

Corollary 2.4 (Recovery of the polynomial spatial freedom)**.** *For $E = \mathcal{P}_N$, $F = \mathcal{P}_{N-1}$, and $D = \mathrm{d}/\mathrm{d}z$, the derivative is onto, $\dim F = N$, and the active compatible family has dimension*

$$\frac{N(N-1)}{2}.$$

Thus the freedom in the polynomial Stokes construction of [20] is the spatial instance of Theorem 2.2, rather than only an analogy based on Gaussian orthogonality.

3 Stokes' theorem on the space–time cylinder

Let Ω be an oriented d-dimensional manifold with boundary and let

$$Z = [0,T] \times \Omega$$

carry the product orientation. Its oriented boundary is

$$\partial Z = (\{T\} \times \Omega) - (\{0\} \times \Omega) \ \cup \ ([0,T] \times \partial\Omega). \tag{9}$$

The first two components will be called the temporal caps and the last one the lateral mantle.

Proposition 3.1 (One identity on the cylinder)**.** *For $\alpha \in \Omega^k(Z)$ and $\beta \in \Omega^{d-k}(Z)$,*

$$\int_Z \mathrm{d}\alpha \wedge \beta + (-1)^k \int_Z \alpha \wedge \mathrm{d}\beta = \int_{\partial Z} \operatorname{tr}\alpha \wedge \operatorname{tr}\beta. \tag{10}$$

The right-hand side splits according to Equation (9) into a final-minus-initial cap term and a lateral boundary term.

Proof. The graded Leibniz rule gives $\mathrm{d}(\alpha\wedge\beta) = \mathrm{d}\alpha\wedge\beta + (-1)^k\alpha\wedge\mathrm{d}\beta$. Apply Stokes' theorem and use the product orientation of Equation (9). □

Corollary 3.2 (Port-Hamiltonian balance as a cylinder law)**.** *Suppose that a distributed port-Hamiltonian system satisfies*

$$\partial_t x = (\mathcal{J} - \mathcal{R})e, \qquad e = \frac{\delta H}{\delta x}, \qquad \mathcal{R} \succeq 0,$$

and that the Green identity for $\mathcal{J}$ is

$$\langle e, \mathcal{J}e\rangle_\Omega = \langle y, u\rangle_{\partial\Omega}.$$

Then

$$H(x(T)) - H(x(0)) = \int_0^T \langle y, u\rangle_{\partial\Omega} \ \mathrm{d}t - \int_0^T \langle e, \mathcal{R}e\rangle_\Omega \ \mathrm{d}t. \tag{11}$$

Thus stored energy on the temporal caps and supplied power through the lateral mantle are components of one oriented boundary balance.

Proof. The variational chain rule and the evolution equation give

$$\frac{\mathrm{d}}{\mathrm{d}t}H(x(t)) = \langle e, (\mathcal{J}-\mathcal{R})e\rangle_\Omega = \langle e, \mathcal{J}e\rangle_\Omega - \langle e, \mathcal{R}e\rangle_\Omega.$$

Use the Green identity and integrate in time. This is the port-Hamiltonian specialization of the cap–mantle split in Proposition 3.1. □

4 Discrete cylinders and the full space–time gauge

4.1 The temporal–spatial total complex

Definition 4.1 (Finite graded Stokes complex)**.** *A finite cochain complex $(C^\bullet, d)$ of formal dimension m is equipped with a graded Stokes pairing if it has bilinear maps*

$$I_k : C^k \times C^{m-k} \to \mathbb{R}$$

and boundary maps B_k such that, for $a \in C^k$ and $b \in C^{m-k-1}$,

$$I_{k+1}(da, b) + (-1)^k I_k(a, db) = B_k(a, b). \tag{12}$$

The boundary maps include the appropriate discrete traces. This is the cochain form of integration by parts; no metric or constitutive operator is included in the definition.

Let $(C_t^\bullet, d_t)$ and $(C_x^\bullet, d_x)$ be finite cochain complexes. Their total complex is

$$C_{\mathrm{st}}^q = \bigoplus_{r+s=q} C_t^r \otimes C_x^s, \tag{13}$$

with, on a homogeneous temporal cochain $a \in C_t^r$,

$$d_{\mathrm{st}}(a \otimes b) = d_t a \otimes b + (-1)^r a \otimes d_x b. \tag{14}$$

In matrices this is

$$D_{\mathrm{st}} = D_t \otimes I + \Gamma_t \otimes D_x, \tag{15}$$

where Γ_t acts by $(-1)^r$ on temporal degree r.

Proposition 4.2 (Complex property and signed product Stokes formula)**.** *The total differential satisfies $d_{\mathrm{st}}^2 = 0$. Assume that the factor complexes have formal dimensions m_t, m_x and carry the pairings of Definition 4.1. For homogeneous simple tensors of complementary bidegrees, define*

$$I_{\mathrm{st}}(a_t \otimes a_x, c_t \otimes c_x) = (-1)^{|a_x||c_t|} I_t(a_t, c_t) I_x(a_x, c_x). \tag{16}$$

Let $|a_t| = r$, $|a_x| = s$. A homogeneous element of total degree $m_t + m_x - r - s - 1$ has the two complementary components

$$b^{(t)} = c_t \otimes c_x, \qquad (|c_t|, |c_x|) = (m_t - r - 1, m_x - s),$$

and

$$b^{(x)} = e_t \otimes e_x, \qquad (|e_t|, |e_x|) = (m_t - r, m_x - s - 1).$$

Then the total boundary form is

$$\begin{aligned} B_{\mathrm{st}}(a, b) = {} & (-1)^{s(m_t - r - 1)} B_t(a_t, c_t) I_x(a_x, c_x) \\ & + (-1)^{m_t + s(m_t - r)} I_t(a_t, e_t) B_x(a_x, e_x). \end{aligned} \tag{17}$$

The first term is the temporal-cap contribution and the second is the spatial-mantle contribution, including the product-orientation sign.

Proof. For a homogeneous simple tensor,

$$d_{\mathrm{st}}^2(a \otimes b) = d_t^2 a \otimes b + (-1)^{r+1} d_t a \otimes d_x b + (-1)^r d_t a \otimes d_x b + a \otimes d_x^2 b = 0.$$

For $b^{(t)}$, only the two temporal derivatives can pair factorwise. Using Equation (16), their sum is

$$(-1)^{s(m_t-r-1)}\big[I_t(d_t a_t, c_t) + (-1)^r I_t(a_t, d_t c_t)\big] I_x(a_x, c_x),$$

which is the first term of Equation (17). For $b^{(x)}$, the sign $(-1)^r$ in the spatial part of Equation (14) combines with the interchange sign in Equation (16). The result is

$$(-1)^{m_t+s(m_t-r)} I_t(a_t, e_t)\big[I_x(d_x a_x, e_x) + (-1)^s I_x(a_x, d_x e_x)\big],$$

which gives the mantle term. Linearity handles sums of bidegrees. This also displays explicitly why the two boundary pieces add while the mixed interior derivatives cancel. □

The affine freedom of the genuine total complex can now be stated without replacing it by a directional gradient. Let m be its formal dimension and fix $q \in \{0, \dots, m-1\}$. Set

$$\mathbb{E}_q = C^q_{\mathrm{st}} \oplus C^{m-q-1}_{\mathrm{st}}, \qquad \mathbb{F}_q = C^{q+1}_{\mathrm{st}} \oplus C^{m-q}_{\mathrm{st}}, \tag{18}$$

and define

$$\mathbb{D}_q(a, b) = (d_{\mathrm{st}} a, d_{\mathrm{st}} b). \tag{19}$$

For $z = (a, b)$ and $\widehat{z} = (\widehat{a}, \widehat{b})$, introduce the symmetric doubled boundary form

$$\mathbf{b}_q(z, \widehat{z}) = B_q(a, \widehat{b}) + B_q(\widehat{a}, b). \tag{20}$$

Theorem 4.3 (Affine classification for the genuine total complex)**.** *Put*

$$U_q = C^q_{\mathrm{st}} / \ker d^q_{\mathrm{st}}, \qquad V_q = C^{m-q-1}_{\mathrm{st}} / \ker d^{m-q-1}_{\mathrm{st}}.$$

The pairing

$$\mathbb{M}_q\big((a, b), (f, g)\big) = I_{q+1}(f, b) + (-1)^q I_q(a, g) \tag{21}$$

is compatible with $\mathbb{D}_q$ and $\mathbf{b}_q$. Without a degree restriction, the active compatible family is affine over

$$\Lambda^2(U_q \oplus V_q)^*. \tag{22}$$

If the pairing is required to preserve complementary degree—so that it pairs C^q_{st} only with dC^{m-q-1}_{st}, and C^{m-q-1}_{st} only with dC^q_{st}—its homogeneous freedom is instead exactly

$$\boxed{\mathrm{Bil}(U_q, V_q; \mathbb{R}) \simeq U_q^* \otimes V_q^*.} \tag{23}$$

Thus cancellations between different bidegrees are incorporated through the actual kernels of the total differential.

Proof. For $z = (a, b)$ and $\widehat{z} = (\widehat{a}, \widehat{b})$, the left-hand side of the symmetric compatibility identity is

$$\begin{aligned}
&\mathbb{M}_q(z, \mathbb{D}_q \widehat{z}) + \mathbb{M}_q(\widehat{z}, \mathbb{D}_q z) \\
&= I_{q+1}(d\widehat{a}, b) + (-1)^q I_q(\widehat{a}, db) + I_{q+1}(da, \widehat{b}) + (-1)^q I_q(a, d\widehat{b}) \\
&= B_q(\widehat{a}, b) + B_q(a, \widehat{b}),
\end{aligned}$$

which is Equation (20). Since $\ker \mathbb{D}_q = \ker d^q_{\mathrm{st}} \oplus \ker d^{m-q-1}_{\mathrm{st}}$, Theorem 2.2 gives Equation (22).

For the degree-preserving subfamily, let a homogeneous correction have off-diagonal components $N_1(a, d\widehat{b})$ and $N_2(b, d\widehat{a})$. After quotienting the two kernels, compatibility with zero boundary requires

$$N_1(u, \widehat{v}) + N_2(v, \widehat{u}) + N_1(\widehat{u}, v) + N_2(\widehat{v}, u) = 0.$$

Taking $\widehat{u} = 0$ and $v = 0$ gives $N_2(\widehat{v}, u) = -N_1(u, \widehat{v})$. Hence one arbitrary bilinear form $\eta \in \mathrm{Bil}(U_q, V_q; \mathbb{R})$ determines both components through

$$N_\eta((a,b), \mathbb{D}_q(\widehat{a}, \widehat{b})) = \eta([a], [\widehat{b}]) - \eta([\widehat{a}], [b]), \tag{24}$$

and every degree-preserving correction is obtained in this way. □

Remark 4.4 (The total and stacked kernels are different)**.** *Take two copies of the complex* $0 \to \mathbb{R} \xrightarrow{1} \mathbb{R} \to 0$*. At total degree one, the total differential is, up to orientation,* $d_{\mathrm{st}}(a,b) = b - a$*; its kernel is the diagonal even though both factor differentials have zero kernel. Therefore the formula* $\ker D_{\mathrm{st}} = K_t \otimes K_x$ *below is not a formula for an arbitrary total-complex differential. It belongs to the stacked one-bidegree map, where the two directional derivatives are retained as separate output components.*

The preceding theorem treats the genuine graded total complex. The finite tensor cells used later employ a related but distinct stacked map. Let $(E_a, F_a, D_a, b_a, M_a)$, $a \in \{t, x\}$, be compatible Stokes data, and let $G_a : E_a \times E_a \to \mathbb{R}$ be symmetric interior pairings. Put

$$E_{\mathrm{st}} = E_t \otimes E_x, \tag{25}$$
$$F_{\mathrm{st}} = (F_t \otimes E_x) \oplus (E_t \otimes F_x), \tag{26}$$
$$D_{\mathrm{st}} z = ((D_t \otimes I)z, (I \otimes D_x)z), \tag{27}$$

where the sign Γ_t is inserted in the second component when the chosen bidegree requires it. Unlike Equation (14), the two directional images in Equation (27) are not added.

Theorem 4.5 (Product representative and cylinder boundary)**.** *There is a compatible product representative* M^0_{st} *defined on simple tensors by*

$$\begin{aligned} &M^0_{\mathrm{st}}(z_t \otimes z_x, (f_t \otimes e_x, e_t \otimes f_x)) \\ &\quad = M_t(z_t, f_t) G_x(z_x, e_x) + G_t(z_t, e_t) M_x(z_x, f_x), \end{aligned} \tag{28}$$

and extended bilinearly. Its boundary form is

$$b_{\mathrm{st}} = b_t \otimes G_x + G_t \otimes b_x. \tag{29}$$

The first summand is supported on the temporal caps and the second on the spatial mantle.

Proof. It suffices to test pure tensors $z = z_t \otimes z_x$ and $w = w_t \otimes w_x$. Substitution of Equation (27) into Equation (28) gives

$$\begin{aligned} &\big[M_t(z_t, D_t w_t) + M_t(w_t, D_t z_t)\big] G_x(z_x, w_x) \\ &\quad + G_t(z_t, w_t)\big[M_x(z_x, D_x w_x) + M_x(w_x, D_x z_x)\big]. \end{aligned}$$

Factor compatibility turns this expression into $b_t(z_t, w_t) G_x(z_x, w_x) + G_t(z_t, w_t) b_x(z_x, w_x)$. Pure tensors span E_{st}, proving the result. In the graded version the Koszul sign in Equation (14) produces the oriented sign in the mantle term. □

4.2 Classification and sector decomposition

Write $K_a = \ker D_a$, $n_a = \dim E_a$, and $k_a = \dim K_a$ for $a \in \{t, x\}$.

Theorem 4.6 (Full active gauge of the stacked cylinder cell)**.** *For Equation (27),*

$$K_{\mathrm{st}} := \ker D_{\mathrm{st}} = K_t \otimes K_x. \tag{30}$$

Consequently

$$r_{\mathrm{st}} := \operatorname{rank} D_{\mathrm{st}} = n_t n_x - k_t k_x. \tag{31}$$

Every active compatible space–time pairing is

$$M^{\omega}_{\mathrm{st}}(z, D_{\mathrm{st}} w) = M^{0}_{\mathrm{st}}(z, D_{\mathrm{st}} w) + \omega([z],[w]), \qquad \omega \in \mathcal{G}_{\mathrm{st}}, \tag{32}$$

where

$$\boxed{\mathcal{G}_{\mathrm{st}} = \Lambda^2 (E_{\mathrm{st}}/K_{\mathrm{st}})^*.} \tag{33}$$

Its dimension is

$$\dim \mathcal{G}_{\mathrm{st}} = \frac{r_{\mathrm{st}}(r_{\mathrm{st}}-1)}{2}. \tag{34}$$

Proof. The kernel of $D_t \otimes I$ is $K_t \otimes E_x$, while the kernel of $I \otimes D_x$ is $E_t \otimes K_x$. Their intersection is $K_t \otimes K_x$, proving Equation (30); rank–nullity gives Equation (31). The product representative in Theorem 4.5 proves existence, and Theorem 2.2 applied to D_{st} gives Equation (32)–Equation (34). □

Proposition 4.7 (Canonical pure–mixed filtration)**.** *Let*

$$Q_{\mathrm{st}} = (E_t \otimes E_x)/(K_t \otimes K_x), \qquad \overline{E}_t = E_t/K_t, \quad \overline{E}_x = E_x/K_x.$$

There is a canonical short exact sequence

$$0 \longrightarrow (\overline{E}_t \otimes K_x) \oplus (K_t \otimes \overline{E}_x) \xrightarrow{J} Q_{\mathrm{st}} \xrightarrow{\pi_{tx}} \overline{E}_t \otimes \overline{E}_x \longrightarrow 0. \tag{35}$$

The subspace on the left contains the pure temporal and pure spatial directions. The quotient on the right is the canonical mixed sector. A choice of complements of K_t and K_x supplies a natural factorized splitting of this filtration. General linear sections of π_{tx} need not arise from such a pair of complements.

Proof. The tensor product of the quotient maps $E_t \to \overline{E}_t$ and $E_x \to \overline{E}_x$ vanishes on $K_t \otimes K_x$, hence induces π_{tx}. Its kernel is the image of $E_t \otimes K_x + K_t \otimes E_x$ modulo $K_t \otimes K_x$. Quotienting the intersection $(E_t \otimes K_x) \cap (K_t \otimes E_x) = K_t \otimes K_x$ identifies this kernel with the direct sum on the left of Equation (35). Surjectivity follows from that of the two factor quotient maps. □

Corollary 4.8 (Temporal, spatial, and mixed sectors)**.** *Choose complements $E_t = K_t \oplus U_t$ and $E_x = K_x \oplus U_x$. Then*

$$E_{\mathrm{st}}/K_{\mathrm{st}} \simeq Q_t \oplus Q_x \oplus Q_{tx}, \tag{36}$$

where

$$Q_t = U_t \otimes K_x, \qquad Q_x = K_t \otimes U_x, \qquad Q_{tx} = U_t \otimes U_x \simeq \overline{E}_t \otimes \overline{E}_x.$$

Accordingly,

$$\mathcal{G}_{\mathrm{st}} \simeq \bigoplus_{a \in \{t,x,tx\}} \Lambda^2 Q_a^* \oplus \bigoplus_{a<b} Q_a^* \otimes Q_b^*. \tag{37}$$

The summands contain pure temporal, pure spatial, mixed, and cross-sector gauges. The total space $\mathcal{G}_{\mathrm{st}}$, the pure subspace, and the mixed quotient in Equation (35) are intrinsic; only the displayed direct-sum splitting depends on the chosen complements.

Proof. Expanding $(K_t \oplus U_t) \otimes (K_x \oplus U_x)$ and quotienting by $K_t \otimes K_x$ gives Equation (36). For a direct sum $Q = Q_1 \oplus Q_2 \oplus Q_3$, the standard exterior-square identity is $\Lambda^2 Q^* = \bigoplus_i \Lambda^2 Q_i^* \oplus \bigoplus_{i<j} Q_i^* \otimes Q_j^*$, which proves Equation (37). □

Proposition 4.9 (Pure-axis embeddings and boundary reduction)**.** *For nonzero $k_t \in K_t$ and $k_x \in K_x$, the maps*

$$\iota_t : E_t/K_t \longrightarrow E_{\mathrm{st}}/K_{\mathrm{st}}, \qquad [z_t] \longmapsto [z_t \otimes k_x],$$
$$\iota_x : E_x/K_x \longrightarrow E_{\mathrm{st}}/K_{\mathrm{st}}, \qquad [z_x] \longmapsto [k_t \otimes z_x]$$

are injective. Pullback by ι_t and ι_x sends a space–time gauge to its pure temporal and pure spatial restrictions. In particular, the time-constant pullback contains the polynomial spatial gauges of [20]. A cap-to-cap temporal method on the full spatial state space is obtained only after eliminating the interior spatial flows; it generally receives contributions from $Q_t \oplus Q_{tx}$. The next sections classify that reduced boundary gauge. A gauge supported entirely on Q_{tx} can be nonzero while both pure-axis restrictions vanish.

Proof. If $z_t \otimes k_x \in K_t \otimes K_x$ and $k_x \neq 0$, then $z_t \in K_t$; hence ι_t is injective. The spatial statement is identical. Alternating forms pull back functorially. Finally a form supported on Q_{tx} vanishes on the images of both pure embeddings. □

4.3 Regular volume-to-cap reduction

The pullbacks in Proposition 4.9 restrict a gauge along pure axes. A numerical time step requires a different operation: the interior equations must first be solved and the resulting active volume field restricted to its cap data. This operation depends on the Dirac relation, the constitutive closure, the boundary causality, and the step size; it is not determined by the Stokes pairing alone.

Let

$$Q_{\mathrm{st}} = E_{\mathrm{st}}/K_{\mathrm{st}}$$

be the active volume space and let Q_∂ be the chosen active cap space. After fixing the four ingredients above, call the reduction *regular* if the interior equations determine a linear reconstruction

$$L_h : Q_\partial \longrightarrow Q_{\mathrm{st}}. \tag{38}$$

The notation permits L_h to depend on the selected constitutive law and on h, but it is held fixed while the following affine gauge reduction is classified.

Theorem 4.10 (Exact visibility sequence for volume-to-cap gauges)**.** *For a regular reconstruction L_h, the homogeneous volume-to-cap reduction is the pullback map*

$$\mathcal{R}_{L_h} : \Lambda^2 Q_{\mathrm{st}}^* \longrightarrow \Lambda^2 Q_\partial^*, \qquad \mathcal{R}_{L_h}(\omega) = L_h^* \omega, \tag{39}$$

where

$$(L_h^*\omega)(\xi, \zeta) = \omega(L_h\xi, L_h\zeta).$$

Put

$$W_h = \operatorname{im} L_h, \qquad K_{\partial,h} = \ker L_h, \qquad r_h = \operatorname{rank} L_h.$$

Then

$$\operatorname{im} \mathcal{R}_{L_h} = \left\{ \eta \in \Lambda^2 Q_\partial^* : \iota_k \eta = 0 \text{ for every } k \in K_{\partial,h} \right\} \simeq \Lambda^2 (Q_\partial / K_{\partial,h})^*, \tag{40}$$

and

$$\ker \mathcal{R}_{L_h} = \left\{ \omega \in \Lambda^2 Q_{\mathrm{st}}^* : \omega|_{W_h \times W_h} = 0 \right\}. \tag{41}$$

Equivalently, restriction to W_h gives the short exact sequence

$$0 \longrightarrow \ker \mathcal{R}_{L_h} \longrightarrow \Lambda^2 Q_{\mathrm{st}}^* \xrightarrow{\mathcal{R}_{L_h}} \Lambda^2 (Q_\partial / K_{\partial,h})^* \longrightarrow 0. \tag{42}$$

Consequently,

$$\dim \operatorname{im} \mathcal{R}_{L_h} = \frac{r_h(r_h-1)}{2}, \tag{43}$$

$$\dim \ker \mathcal{R}_{L_h} = \binom{\dim Q_{\mathrm{st}}}{2} - \binom{r_h}{2}. \tag{44}$$

If L_h is injective, every cap gauge has a volume extension. If L_h is surjective, no nonzero volume gauge is invisible on all cap data. In particular, an isomorphism L_h induces an isomorphism of gauge spaces.

Proof. If $k \in K_{\partial,h}$, then $(L_h^*\omega)(k,\xi) = \omega(0, L_h\xi) = 0$, proving the inclusion from left to right in Equation (40). Conversely, every η satisfying the annihilation condition descends uniquely to an alternating form on $Q_\partial/K_{\partial,h}$. The induced map

$$\overline{L}_h : Q_\partial/K_{\partial,h} \longrightarrow W_h$$

is an isomorphism. Transport η to W_h by this isomorphism and extend the resulting alternating form from W_h to Q_{st} using any linear complement of W_h. Its pullback is η, so Equation (40) and surjectivity in Equation (42) follow.

The pullback vanishes exactly when ω vanishes on every pair of vectors in W_h, which is Equation (41). Since $\dim W_h = r_h$, the two dimension formulas follow. Injectivity of L_h means $K_{\partial,h} = 0$, while surjectivity means $W_h = Q_{\mathrm{st}}$; the final claims are immediate. □

Corollary 4.11 (Affine reduction and gauge-invariant power)**.** *Assume that a reference compatible volume pairing reduces, under the same fixed reconstruction, to a cap pairing $m_{\partial,0}$. Then the full reduced family is*

$$m_{\partial,\omega} = m_{\partial,0} + L_h^*\omega. \tag{45}$$

For every cap increment ξ,

$$(L_h^*\omega)(\xi,\xi) = 0.$$

Thus the visible gauge changes the cap method but not its contracted power identity.

Corollary 4.12 (Intrinsic reduction of the mixed quotient)**.** *Let $Q_{tx} = \overline{E}_t \otimes \overline{E}_x$ and let $\pi_{tx} : Q_{\mathrm{st}} \to Q_{tx}$ be the canonical quotient map in Equation (35). Pullback embeds the canonical mixed gauges as $\pi_{tx}^*\Lambda^2 Q_{tx}^* \subset \Lambda^2 Q_{\mathrm{st}}^*$, and*

$$\boxed{\mathcal{R}_{L_h} \circ \pi_{tx}^* = (\pi_{tx} L_h)^*.} \tag{46}$$

If $r_{tx,h} = \operatorname{rank}(\pi_{tx}L_h)$, then the visible canonical mixed family has dimension

$$\dim \operatorname{im}(\mathcal{R}_{L_h} \circ \pi_{tx}^*) = \binom{r_{tx,h}}{2}. \tag{47}$$

Its kernel consists exactly of the forms on Q_{tx} that vanish on $\operatorname{im}(\pi_{tx}L_h) \times \operatorname{im}(\pi_{tx}L_h)$, and has dimension

$$\binom{\dim Q_{tx}}{2} - \binom{r_{tx,h}}{2}.$$

In particular, there exists a nonzero canonical mixed gauge whose cap pullback is nonzero if and only if

$$\operatorname{rank}(\pi_{tx}L_h) \geq 2. \tag{48}$$

A prescribed nonzero mixed form may nevertheless belong to the kernel.

Proof. Functoriality gives $L_h^* \pi_{tx}^* = (\pi_{tx} L_h)^*$. Apply Theorem 4.10 to the single map $\pi_{tx} L_h$. The image, kernel, and dimension statements follow verbatim and require no choice of tensor complements. □

Proposition 4.13 (Schur realization of the reconstruction)**.** *Suppose a linear Dirac–constitutive cell has been partitioned into interior variables z_i and cap variables ξ, and its interior equations are*

$$K_{ii}(h) z_i + K_{i\partial}(h) \xi = 0. \tag{49}$$

If $K_{ii}(h)$ is invertible, the regular reconstruction is

$$L_h \xi = \begin{bmatrix} -K_{ii}(h)^{-1} K_{i\partial}(h) \xi \\ \xi \end{bmatrix}. \tag{50}$$

In coordinates, a volume gauge matrix $\Omega_{\mathrm{st}}^\top = -\Omega_{\mathrm{st}}$ reduces to

$$\Omega_\partial = L_h^\top \Omega_{\mathrm{st}} L_h. \tag{51}$$

Proof. Equation Equation (50) is the unique solution of Equation (49); Equation (51) is the coordinate form of Equation (39). □

Theorem 4.14 (Trace-generated cell and exact raw realization)**.** *Let X be the active cap state space and Z_h a raw cell space. Let $T_h : Z_h \to X$ be onto, let $Q : X \to X^*$ be symmetric, and let $J : X^* \to X$ be skew. For an alternating cap map $S_h : X \to X^*$, put*

$$\Omega_h = T_h^* S_h T_h : Z_h \longrightarrow Z_h^*. \tag{52}$$

It descends to $\overline{\Omega}_h$ on $\overline{Z}_h = Z_h / \ker T_h$. Given x_0, consider the weak cell equations for $([z], e, x_1) \in \overline{Z}_h \times X^* \times X$:

$$\overline{T}_h[z] = x_1 - x_0, \tag{53}$$

$$x_1 - x_0 = hJe, \tag{54}$$

$$\overline{T}_h^*(e - Q\overline{x}) + \overline{\Omega}_h[z] = 0, \qquad \overline{x} = \tfrac{1}{2}(x_0 + x_1). \tag{55}$$

Then the cap closure encoded by Ω_h is recovered exactly by elimination:

$$e = Q\overline{x} - S_h(x_1 - x_0) \tag{56}$$

and its exact Schur complement is

$$\boxed{(I + hJS_h - \tfrac{h}{2}JQ)x_1 = (I + hJS_h + \tfrac{h}{2}JQ)x_0.} \tag{57}$$

Equation Equation (57) is always a linear relation. If

$$I + hJS_h - \tfrac{h}{2}JQ \quad \text{is invertible,} \tag{58}$$

it defines a unique map from x_0 to x_1. The physical variables (e, x_1) are independent of the representative of $[z]$ and of the right inverse used to coordinatize the quotient.

In Euclidean coordinates, let the columns of N_h form an orthonormal basis of $\ker T_h$. A square raw assembly, with a multiplier fixing the null component, is

$$\begin{bmatrix} T_h & 0 & -I & 0 \\ 0 & -hJ & I & 0 \\ \Omega_h & T_h^* & -\frac{1}{2}T_h^*Q & N_h \\ N_h^* & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} z \\ e \\ x_1 \\ \lambda \end{bmatrix} = \begin{bmatrix} -I \\ I \\ \frac{1}{2}T_h^*Q \\ 0 \end{bmatrix} x_0. \tag{59}$$

Under Equation (58), this square system is invertible and its Schur map from x_0 to x_1 is exactly Equation (57). Thus the prescribed cap gauge is not attached as an additional reduced equation: it is encoded by the trace-generated volume form and recovered exactly from the raw cell.

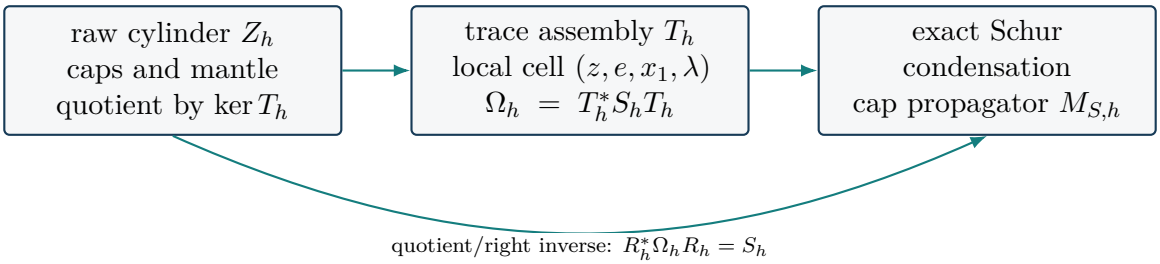


Figure 2: Exact raw-cell realization of a prescribed cap gauge. The lower identity is independent of the chosen section; the upper path shows that the gauge encoded in $\Omega_h = T_h^* S_h T_h$ is recovered exactly by elimination. No claim of selecting S_h from independent interior dynamics is made here.

Proof. Because T_h is onto, the induced map $\overline{T}_h : \overline{Z}_h \to X$ is an isomorphism and $\overline{T}_h^*$ is injective. Using Equations (52) and (53), the third cell equation becomes

$$\overline{T}_h^*(e - Q\overline{x} + S_h(x_1 - x_0)) = 0.$$

Injectivity gives Equation (56). Substitution into Equation (54) gives

$$x_1 - x_0 = hJ(Q\overline{x} - S_h(x_1 - x_0)),$$

which rearranges to Equation (57). If $R_h : X \to Z_h$ is any right inverse, then

$$R_h^* \Omega_h R_h = (T_h R_h)^* S_h (T_h R_h) = S_h.$$

Two right inverses differ by a map into $\ker T_h$, on which Ω_h vanishes. Finally, the last row of Equation (59) selects the orthogonal representative; the third row is the unquotiented weak equation. Eliminating (z, e, λ) reproduces the preceding calculation. Under Equation (58), a homogeneous right-hand side first gives $x_1 = e = 0$ from the condensed equations, then $z = 0$ from $T_h z = 0$ and $N_h^* z = 0$, and finally $\lambda = 0$. The square raw system is therefore invertible. □

Remark 4.15 (Singular interior problems)**.** *If $K_{ii}(h)$ is singular, the interior solution may be nonunique or subject to compatibility constraints. The reduction is then a linear relation rather than a map L_h, and composition of Dirac relations or presymplectic reduction is required. Likewise, if changing the volume gauge changes the interior solve, one obtains L_h^ω and the global map $\omega \mapsto (L_h^\omega)^* \omega$ need not be affine. Theorem 4.10 classifies the exact gauge freedom within each fixed regular reduction chart; it does not hide these hypotheses.*

Proposition 4.16 (Tangent reduction when the reconstruction depends on the gauge)**.** *Suppose $\omega \mapsto L_h^\omega$ is differentiable and define*

$$\mathcal{F}_h(\omega) = (L_h^\omega)^* \omega.$$

For $\xi, \zeta \in Q_\partial$,

$$\begin{aligned} D\mathcal{F}_h(\omega)[\delta\omega](\xi, \zeta) = {} & \delta\omega(L_h^\omega \xi, L_h^\omega \zeta) \\ & + \omega(DL_h^\omega[\delta\omega]\xi, L_h^\omega \zeta) \\ & + \omega(L_h^\omega \xi, DL_h^\omega[\delta\omega]\zeta). \end{aligned} \tag{60}$$

At the zero gauge,

$$D\mathcal{F}_h(0)[\delta\omega] = L_h^{0*} \delta\omega. \tag{61}$$

Thus the fixed-reconstruction pullback theorem is also the exact tangent model of a gauge-dependent nonlinear reduction. For the trace-generated family in Theorem 4.14, the stronger finite identity holds because the quotient trace itself is gauge independent.

Proof. Differentiate $\omega(L_h^\omega \xi, L_h^\omega \zeta)$ by the product rule. At $\omega = 0$, the two terms containing the undifferentiated form vanish. □

5 The complete two-point temporal gauge

We now pass to the cap-to-cap problem obtained after eliminating interior spatial flows, while keeping the spatial state space arbitrary. In the language of Theorem 4.10, its gauge belongs to the image of $\mathcal{R}_{L_h}$ and can therefore contain the trace of mixed space–time sectors precisely under Equation (48). This is also the smallest setting in which the distinction between gauge-invariant power and gauge-dependent symplecticity is visible.

5.1 Temporal Stokes data

Let $V = \mathbb{R}^n$, let $Q = Q^\top$, and define

$$H(x) = \tfrac{1}{2} x^\top Q x, \qquad z = (x_0, x_1) \in E_t = V \oplus V, \qquad D_t z = x_1 - x_0 =: \Delta x.$$

The polarized temporal boundary form is

$$b_Q(z, w) = x_1^\top Q y_1 - x_0^\top Q y_0, \qquad w = (y_0, y_1). \tag{62}$$

Its restriction to $K_t = \ker D_t = \{(x, x) : x \in V\}$ vanishes. Hence Theorem 2.2 applies.

A linear two-point effort is a linear map $\bar{e} : E_t \to V^*$. It defines

$$M_{\bar{e}}(z, v) = \bar{e}(z)[v].$$

Taking $w = z$ in Equation (4) shows that compatibility is exactly the polarized form of

$$\bar{e}(x_0, x_1)[\Delta x] = H(x_1) - H(x_0). \tag{63}$$

Theorem 5.1 (Complete linear temporal gauge)**.** *The linear efforts satisfying Equation (63) for all x_0, x_1 are precisely*

$$\bar{e}_{\mathsf{S}}(x_0, x_1) = Q\bar{x} + \iota_{\Delta x}\mathsf{S}, \qquad \bar{x} = \tfrac{1}{2}(x_0 + x_1), \quad \mathsf{S} \in \Lambda^2 V^*. \tag{64}$$

In coordinates, if $S^\top = -S$ represents S, then

$$\boxed{\bar{e}_S = Q\bar{x} - S\Delta x}. \tag{65}$$

The affine family has dimension $n(n-1)/2$.

Proof. The midpoint effort $Q\bar{x}$ is compatible because

$$(Q\bar{x})^\top \Delta x = \tfrac{1}{2}(x_1 + x_0)^\top Q(x_1 - x_0) = H(x_1) - H(x_0).$$

Moreover, $E_t/K_t \simeq V$ through D_t. Applying Theorem 2.2, the difference from the midpoint pairing is an arbitrary $\mathsf{S} \in \Lambda^2 V^*$ evaluated on $(\Delta x, v)$. Since $\mathsf{S}(\Delta x, v) = (-S\Delta x)^\top v$, this is Equation (65). Alternation makes its contraction with Δx vanish. □

5.2 Gauge-invariant power balance

Consider the port-Hamiltonian step

$$\frac{\Delta x}{h} = (J - R)\bar{e}_S + G\bar{u}, \qquad \bar{y} = G^\top \bar{e}_S, \tag{66}$$

where $J^\top = -J$ and $R^\top = R \succeq 0$.

Corollary 5.2 (Exact gauge-invariant power law)**.** *Every $S^\top = -S$ gives the exact identity*

$$H(x_1) - H(x_0) = h\,\bar{y}^\top \bar{u} - h\,\bar{e}_S^\top R \bar{e}_S. \tag{67}$$

For $R = 0$, the discrete boundary supply is exact. For $R \succeq 0$, the step is passive.

Proof. Pair Equation (66) with $\bar{e}_S$, use Equation (63), skewness of J, and the definition of $\bar{y}$. □

5.3 Constitutively shifted Cayley coordinates

The classical Cayley transform gives rational coordinates between a Lie algebra and an open cell of its matrix group [19, 24]. The point here is not to rename that transform, but to identify the *constitutively shifted* variable $I + hJS$ selected by the discrete chain rule. At fixed h, the cap gauges thereby parametrize the energy-orthogonal maps having no eigenvalue $+1$. This convention differs from the common Cayley cell that excludes -1: the group identity is not in the present cell, and the zero gauge represents the midpoint map rather than the identity.

Theorem 5.3 (Gauge coordinate on the no-$+1$ Cayley cell of $O(Q)$)**.** *Assume that $J^\top = -J$ and $Q = Q^\top$ are nonsingular, put $A = JQ$, $\Omega = J^{-1}$, and fix $h \neq 0$. For an alternating matrix S, define*

$$B_\pm(S) = I + hJS \pm \frac{h}{2}A, \qquad M_S = B_-(S)^{-1}B_+(S), \tag{68}$$

whenever $B_-(S)$ is invertible. Then

$$\Big\{S : S^\top = -S,\ B_-(S) \text{ invertible}\Big\} \xrightarrow{S \mapsto M_S} \Big\{M : M^\top Q M = Q,\ M - I \text{ invertible}\Big\} \tag{69}$$

is a bijection, denoted by Φ_h. Its inverse is

$$R_M = (M + I)(M - I)^{-1}, \tag{70}$$

$$C_M = \frac{h}{2}AR_M, \qquad S_M = \frac{1}{h}\Omega(C_M - I). \tag{71}$$

Proof. The step equations are $\Delta x = hJ(Q\bar{x} - S\Delta x)$, so the discrete chain rule gives $x_1^\top Q x_1 = x_0^\top Q x_0$. Hence every well-defined M_S is Q-orthogonal. Moreover,

$$M_S - I = B_-(S)^{-1}hA,$$

which is invertible because A is.

Conversely, let $M^\top Q M = Q$ and suppose $M - I$ is invertible. The Cayley ratio in Equation (70) is Q-skew:

$$R_M^\top Q + Q R_M = 0. \tag{72}$$

Indeed, multiplying the left-hand side by $(M - I)^\top$ and $M - I$ reduces it to $2(M^\top Q M - Q)$. Since $A^\top \Omega = -Q$ and $\Omega A = Q$,

$$C_M^\top \Omega = -\frac{h}{2}R_M^\top Q = \frac{h}{2}QR_M = \Omega C_M.$$

It follows that $S_M^\top = -S_M$. Finally,

$$C_M - \frac{h}{2}A = hA(M-I)^{-1}, \qquad C_M + \frac{h}{2}A = hAM(M-I)^{-1},$$

so $B_-(S_M)$ is invertible and $B_-(S_M)^{-1}B_+(S_M) = M$. Solving $B_-(S)M_S = B_+(S)$ for $I+hJS$ gives C_{M_S}, proving that the two constructions are inverse. □

Corollary 5.4 (The exact-flow gauge)**.** *Where $e^{hA} - I$ is invertible, the exact Hamiltonian flow is represented by*

$$S_h^{\text{exact}} = \frac{1}{h}\Omega\left[\frac{hA}{2}\coth\left(\frac{hA}{2}\right) - I\right]. \tag{73}$$

The matrix function in brackets is regular at the origin and

$$S_h^{\text{exact}} = \frac{h}{12}\Omega A^2 - \frac{h^3}{720}\Omega A^4 + O(h^5). \tag{74}$$

Proof. For $M = e^{hA}$, functional calculus gives $(M+I)(M-I)^{-1} = \coth(hA/2)$. Apply Equation (71); the series follows from the Bernoulli expansion of $(z/2)\coth(z/2)$. □

Remark 5.5 (Stable evaluation and chart conditioning)**.** *The inverse formula Equation (70) is an exact algebraic identity, but evaluating it from a rounded matrix M is ill-conditioned as $M \to I$: both the subtraction $M - I$ and its inverse amplify perturbations. This is intrinsic to recovering a finite gauge from the chosen group coordinate near the excluded identity. It is not necessary when the map is known analytically. For the exact flow one evaluates $(z/2)\coth(z/2) - 1$ by its Bernoulli series or a cancellation-free matrix-function algorithm. For Gauss–s, one evaluates Equation (99) directly from the even and odd Padé polynomials, without forming either $R_s - I$ or the inverse Cayley ratio. For the exact flow, the chart has a genuine resonance whenever e^{hA} has eigenvalue $+1$, equivalently when some spectral value satisfies $h\lambda \in 2\pi i\mathbb{Z}$. The local Bernoulli series has radius 2π, set by the nearest nonzero poles of $(z/2)\coth(z/2)$; away from the local disk one should use a Schur–Parlett or rational matrix-function evaluation and test the distance to these resonances. Since $S_h^{\text{exact}} = O(h)$, a rounded evaluation of $M - I$ generically gives a* relative *inverse-chart error of size $O(\epsilon_{\text{mach}}/h^3)$. The scaled values in Table 8 confirm this law. The numerical tests in Table 8 separate this stable forward evaluation from the condition-sensitive inverse chart.*

5.4 Canonical selectors

Proposition 5.6 (Symmetry, localization, and order)**.** *Within Equation (65):*

1. *endpoint symmetry $\bar{e}(x_1, x_0) = \bar{e}(x_0, x_1)$ holds if and only if $S = 0$;*
2. *dependence on the midpoint alone holds if and only if $S = 0$;*
3. *if Q is nonsingular, localization as $\bar{e} = Q((1-\theta)x_0 + \theta x_1)$, with a fixed θ, holds if and only if $\theta = \frac{1}{2}$ and $S = 0$;*
4. *for the closed equation $\dot{x} = JQx$, if both J and Q are nonsingular, the associated one-step map is second order if and only if $S = 0$. Every nonzero fixed S is first order.*

Proof. Interchanging the endpoints changes Δx to $-\Delta x$, proving the first two claims. Point localization would require $-S = (\theta - \frac{1}{2})Q$. Its left-hand side is skew and its right-hand side symmetric, so both vanish.

For the order statement put $A = JQ$. The closed method can be written

$$\Delta x = hF_S\bar{x}, \qquad F_S = (I + hJS)^{-1}A.$$

Its map is the Cayley transform in Equation (68), equivalently $M_S = (I - hF_S/2)^{-1}(I + hF_S/2)$. Expansion gives

$$M_S = I + hA + h^2\left(\tfrac{1}{2}A^2 - JSA\right) + O(h^3). \tag{75}$$

The exact flow has coefficient $A^2/2$. Under the stated invertibility assumptions, $JSA = 0$ implies $S = 0$. □

Remark 5.7 (Necessary hypotheses)**.** *If Q is singular, a nonzero S may satisfy $SA = 0$, so the unqualified claim that every nonzero gauge is first order is false. The hypotheses in Proposition 5.6 are therefore substantive.*

6 The exact symplectic gauge locus

Assume in this section that J is invertible, put $\Omega = J^{-1}$, and keep $Q = Q^\top$. The continuous matrix $A = JQ$ is Hamiltonian:

$$A^\top\Omega + \Omega A = 0.$$

Theorem 6.1 (Symplectic-locus criterion)**.** *Assume that $I + hJS$ and $B_-(S)$ are invertible. Then the map M_S is Ω-symplectic if and only if*

$$\boxed{SJQ = QJS} \qquad \Longleftrightarrow \qquad A^\top S + SA = 0. \tag{76}$$

Thus the symplectic gauges form the linear space

$$\mathcal{L} = \{S \in \Lambda^2 V^* : A^\top S + SA = 0\}$$

of alternating two-forms invariant under the continuous flow.

Proof. A Cayley transform of F_S is Ω-symplectic exactly when $F_S^\top\Omega + \Omega F_S = 0$. Since

$$\Omega F_S = (I + hSJ)^{-1}Q,$$

this condition is the symmetry of $(I + hSJ)^{-1}Q$. Equivalently,

$$(I + hSJ)^{-1}Q = Q(I + hJS)^{-1}.$$

Multiplication on the left and right yields $SJQ = QJS$. Finally, $A^\top = -QJ$, so this is equivalent to $A^\top S + SA = 0$. □

Corollary 6.2 (The symplectic intersection of the Cayley chart)**.** *Under the hypotheses of Theorem 5.3, the bijection Equation (69) restricts to*

$$\mathcal{L} \cap \operatorname{dom}\Phi_h \longleftrightarrow \{M \in O(Q) \cap Sp(\Omega) : M - I \text{ invertible}\}.$$

Thus the centralizer equation $A^\top S + SA = 0$ is precisely the gauge-coordinate description of the intersection between energy orthogonality and symplecticity.

Theorem 6.3 (Exact symplectic defect)**.** *Assume $I + hJS$ is invertible, put $F_S = (I + hJS)^{-1}A$, and let $D_S = I - \frac{h}{2}F_S$. Then*

$$D_S^\top(M_S^\top\Omega M_S - \Omega)D_S = h(F_S^\top\Omega + \Omega F_S), \tag{77}$$

$$F_S^\top\Omega + \Omega F_S = -h(I + hSJ)^{-1}(SJQ - QJS)(I + hJS)^{-1}. \tag{78}$$

Hence an off-locus gauge has a one-step symplectic defect of order h^2, with leading coefficient controlled by $SJQ - QJS$.

Proof. Writing $N_S = I + \frac{h}{2}F_S$, commutativity of N_S and D_S gives $M_S D_S = N_S$. Therefore

$$D_S^\top (M_S^\top \Omega M_S - \Omega) D_S = N_S^\top \Omega N_S - D_S^\top \Omega D_S,$$

which expands to Equation (77). Direct calculation gives

$$F_S^\top \Omega + \Omega F_S = (I + hSJ)^{-1} Q - Q(I + hJS)^{-1}.$$

Multiplying the middle difference by its two denominators proves Equation (78). □

Theorem 6.4 (Centralizer representation and generic dimension)**.** *Let $n = 2m$, $C = JS$, and define the Ω-adjoint by $C^\dagger = \Omega^{-1} C^\top \Omega$. Then*

$$S \in \mathcal{L} \quad \Longleftrightarrow \quad [C, A] = 0, \qquad C^\dagger = C, \qquad S = \Omega C. \tag{79}$$

If A is nonderogatory, i.e. its minimal polynomial has degree $2m$, then

$$\mathcal{L} = \operatorname{span}\{\Omega, \Omega A^2, \ldots, \Omega A^{2m-2}\}, \qquad \dim \mathcal{L} = m. \tag{80}$$

Proof. Substituting $S = \Omega C$ in Equation (76), and using $Q = \Omega A$ and $J\Omega = I$, gives $CA = AC$. Moreover, $(\Omega C)^\top = -\Omega C$ is equivalent to $C^\top \Omega = \Omega C$, namely $C^\dagger = C$.

If A is nonderogatory, its centralizer consists of $p(A)$, $\deg p < 2m$. Since $A^\dagger = -A$,

$$p(A)^\dagger = p(-A).$$

The self-adjoint condition implies $p(-A) = p(A)$. The polynomial $p(-z) - p(z)$ has degree below the minimal-polynomial degree and annihilates A, hence it is zero. Thus p is even. The m even powers in Equation (80) are independent by the same minimal-polynomial argument. □

Corollary 6.5 (Two dimensions and time reparametrization)**.** *The form Ω always belongs to $\mathcal{L}$. If $S = \sigma\Omega$, then the method is the midpoint rule with effective step*

$$h_{\text{eff}} = \frac{h}{1 + h\sigma}.$$

For $n = 2$, every skew S is a multiple of Ω; consequently the full gauge family is symplectic, although only $S = 0$ is second order for a fixed nondegenerate system.

Proof. For $S = \sigma\Omega$, $JS = \sigma I$, and the step equation becomes $(1 + h\sigma)\Delta x = hA\bar{x}$. In two dimensions $\Lambda^2 V^*$ is one-dimensional. □

Example 6.6 (Two uncoupled modes)**.** *Let $J = \operatorname{diag}(J_2, J_2)$, $J_2 = \left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right]$, and $Q = \operatorname{diag}(\omega_1 I_2, \omega_2 I_2)$, with $\omega_1 \neq \omega_2 > 0$. Then*

$$\mathcal{L} = \{\operatorname{diag}(\sigma_1 J_2, \sigma_2 J_2) : \sigma_1, \sigma_2 \in \mathbb{R}\}.$$

Every cross-mode skew block lies outside $\mathcal{L}$ unless it satisfies a frequency-dependent degeneracy. This four-dimensional example is the smallest setting in which exact energy and symplecticity are genuinely separated by the gauge.

Corollary 6.7 (Semidiscrete wave equation)**.** *Let a wave semidiscretization with N configuration degrees of freedom have*

$$J = \begin{bmatrix} 0 & I_N \\ -I_N & 0 \end{bmatrix}, \qquad Q = \begin{bmatrix} K & 0 \\ 0 & I_N \end{bmatrix}, \qquad K = K^\top \succ 0.$$

If K has N distinct eigenvalues, then the state dimension is $2N$ and

$$\dim \mathcal{L} = N, \qquad \mathcal{L} = \operatorname{span}\{\Omega, \Omega A^2, \ldots, \Omega A^{2N-2}\}.$$

If eigenvalues are repeated, the locus can be larger and its dimension is not determined by N alone.

Proof. The Hamiltonian matrix $A = \left[\begin{smallmatrix} 0 & I_N \\ -K & 0 \end{smallmatrix}\right]$ has eigenvalues $\pm i\sqrt{\lambda_j(K)}$. Under the simple-spectrum hypothesis they are all distinct, so A is nonderogatory. Theorem 6.4 applies with $m = N$. Repeated eigenvalues enlarge the centralizer in general. □

7 Nonlinear Hamiltonians and the universal segment rule

Let $H \in C^1(V, \mathbb{R})$. A two-point map $\bar{\nabla} H : V \times V \to V^*$ is a discrete gradient if

$$\bar{\nabla} H(x_0, x_1)[x_1 - x_0] = H(x_1) - H(x_0), \tag{81}$$

with the consistency condition $\bar{\nabla} H(x, x) = \mathrm{d}H(x)$.

Proposition 7.1 (Pointwise nonlinear gauge)**.** *For a fixed pair $x_0 \neq x_1$, if g_0 is one compatible covector, then every compatible covector is*

$$g_0 + r, \qquad r \in \mathrm{Ann}(\Delta x) = \{r \in V^* : r[\Delta x] = 0\}.$$

The affine direction has dimension $n - 1$. Moreover every such r can be written $r = \iota_{\Delta x}\mathsf{S}$ for some $\mathsf{S} \in \Lambda^2 V^*$, although this representation is nonunique for $n > 2$.

Proof. The first assertion is immediate from subtracting two copies of Equation (81). Choose $\alpha \in V^*$ with $\alpha[\Delta x] = 1$. Since $r[\Delta x] = 0$,

$$\iota_{\Delta x}(\alpha \wedge r) = r.$$

□

The average vector field (AVF) discrete gradient is

$$\bar{\nabla}_{\mathrm{AVF}} H(x_0, x_1) = \int_0^1 \mathrm{d}H(x_0 + \tau \Delta x)\ \mathrm{d}\tau. \tag{82}$$

Theorem 7.2 (Uniqueness of the universal segment measure)**.** *Let μ be a finite signed Borel measure on $[0, 1]$, independent of H, x_0, x_1, and set*

$$\bar{\nabla}_{\mu} H(x_0, x_1) = \int_0^1 \mathrm{d}H(x_0 + \tau \Delta x)\ \mathrm{d}\mu(\tau).$$

This formula satisfies the discrete chain rule for every C^1 Hamiltonian and every segment if and only if μ is Lebesgue measure. Thus AVF is the unique universal discrete gradient in this segment-measure class.

Proof. Sufficiency is the fundamental theorem of calculus. For necessity, restrict to a one-dimensional affine line and write $g(\tau) = H(x_0 + \tau \Delta x)$. Compatibility says

$$\int_0^1 g'(\tau)\ \mathrm{d}\mu(\tau) = g(1) - g(0) = \int_0^1 g'(\tau)\ \mathrm{d}\tau$$

for every $g \in C^1[0, 1]$. Every continuous function is the derivative of such a g; hence the two measures have equal integrals against all continuous functions. The uniqueness part of the Riesz representation theorem gives $\mu = \mathrm{d}\tau$. □

Corollary 7.3. *For quadratic H, AVF equals $Q\bar{x}$, the zero-gauge effort of Theorem 5.1.*

Remark 7.4 (What is and is not selected)**.** *Symmetry alone does not select AVF: the Gonzalez discrete gradient is also symmetric [13]. AVF is affine-covariant because it integrates a covector field along an affine segment; the standard Gonzalez formula uses a chosen inner product through* $\|\Delta x\|^{-2}$*. The uniqueness in Theorem 7.2 is therefore precise but deliberately restricted to universal segment-measure formulas. It is consistent with the established order theory and structural characterization of AVF [26, 10].*

Remark 7.5 (Symplecticity versus nonlinear energy preservation)**.** *No universal claim that the nonlinear symplectic gauge locus is empty is made here. Classical no-go results require hypotheses concerning symplecticity, exact energy preservation, consistency, and the absence of additional first integrals. They imply, roughly, that a method satisfying both properties for a generic nonintegrable Hamiltonian must coincide with an exact-flow reparametrization [11, 14]. This is a restriction on method classes, not a proof that every particular nonlinear problem has an empty locus.*

8 Multiple stages: classification for invertible stage matrices

8.1 Bilinear chain-rule pairings

Let $\mathcal{A} \in \mathbb{R}^{s\times s}$ be a Runge–Kutta matrix, let $b \in \mathbb{R}^s$, and let

$$X = \mathbf{1} \otimes x_0 + h(\mathcal{A} \otimes I)F, \qquad x_1 = x_0 + h(b^\top \otimes I)F.$$

For a quadratic Hamiltonian, let $E = (I_s \otimes Q)X$. A matrix $P \in \mathbb{R}^{s\times s}$ defines the stage pairing

$$\langle\!\langle E, F\rangle\!\rangle_P = \sum_{i,j=1}^{s} P_{ij} e_i^\top f_j.$$

Proposition 8.1 (Stage chain-rule conditions)**.** *The identity*

$$H(x_1) - H(x_0) = h\langle\!\langle E, F\rangle\!\rangle_P$$

holds for all x_0, F *and all symmetric* Q *if and only if*

$$\boxed{\mathbf{1}^\top P = b^\top, \qquad \mathcal{A}^\top P + P^\top \mathcal{A} = bb^\top.} \tag{83}$$

Proof. Substitution gives

$$h\langle\!\langle E, F\rangle\!\rangle_P = h\, x_0^\top Q(\mathbf{1}^\top P \otimes I)F + h^2 F^\top(\mathcal{A}^\top P \otimes Q)F.$$

Only the symmetric part of $\mathcal{A}^\top P$ contributes. Comparing this expression with

$$\tfrac{1}{2}(x_0 + h(b^\top \otimes I)F)^\top Q(x_0 + h(b^\top \otimes I)F) - \tfrac{1}{2}x_0^\top Q x_0$$

for arbitrary data yields Equation (83). □

Theorem 8.2 (All compatible stage pairings for invertible $\mathcal{A}$)**.** *Assume that* $\mathcal{A}$ *is invertible. Let*

$$v = \mathcal{A}^{-1}\mathbf{1}, \qquad \beta = b^\top v.$$

A compatible P *exists if and only if*

$$\beta \in \{0, 2\}. \tag{84}$$

When it exists, every solution is

$$P = \mathcal{A}^{-\top}\left(\tfrac{1}{2}bb^\top + K\right), \qquad K^\top = -K, \tag{85}$$

where

$$v^\top K = (1 - \tfrac{1}{2}\beta)b^\top. \tag{86}$$

The solution set is affine of dimension

$$\frac{(s-1)(s-2)}{2}. \tag{87}$$

Equivalently, if $R(z) = 1 + z\, b^\top (I - z\mathcal{A})^{-1}\mathbf{1}$ is the stability function, existence holds exactly when $|R(\infty)| = 1$.

Proof. Put $X = \mathcal{A}^\top P$. The second equation in Equation (83) says

$$X = \tfrac{1}{2} b b^\top + K, \qquad K^\top = -K,$$

which gives Equation (85). The first equation becomes Equation (86). For every skew K, $v^\top K v = 0$. Thus a necessary condition is

$$(1 - \tfrac{1}{2}\beta)b^\top v = (1 - \tfrac{1}{2}\beta)\beta = 0,$$

which is Equation (84). It is sufficient: every row $w^\top$ with $w^\top v = 0$ can be represented as $v^\top K = w^\top$, for example

$$K = \frac{v w^\top - w v^\top}{v^\top v}.$$

The linear map $K \mapsto v^\top K$ from skew matrices has image $v^\perp$, hence rank $s - 1$. Its kernel, and therefore the affine solution family, has dimension

$$\frac{s(s-1)}{2} - (s-1) = \frac{(s-1)(s-2)}{2}.$$

Finally $R(\infty) = 1 - b^\top \mathcal{A}^{-1}\mathbf{1} = 1 - \beta$. □

Remark 8.3 (Singular stage matrices)**.** *Proposition 8.1 remains the complete solvability system when $\mathcal{A}$ is singular, but Theorem 8.2 does not classify that case. It is nonempty: for the trapezoidal rule*

$$\mathcal{A} = \begin{bmatrix} 0 & 0 \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}, \qquad b = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}, \qquad P = \frac{1}{4}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

satisfies Equation (83). A full singular classification is the finite-dimensional range problem for

$$\Phi(P) = (P^\top \mathbf{1},\, \mathcal{A}^\top P + P^\top \mathcal{A});$$

it is deliberately not claimed here.

Remark 8.4 (Nondegeneracy and power skewness)**.** *Existence of P does not imply* $\det P \neq 0$. *Nor does the chain rule alone make the pointwise stage operator $I_s \otimes J$ skew for the P-pairing. For pointwise Hamiltonian stages $f_i = Je_i$, the contraction vanishes automatically when $P = P^\top$. These are separate requirements. The local Gauss pairing below has $P = \mathrm{diag}(b) \succ 0$, so both issues disappear.*

8.2 Locality, the Cooper condition, and Gauss rigidity

Take distinct real collocation nodes $c_1, \dots, c_s$, with Lagrange polynomials ℓ_j, and set

$$a_{ij} = \int_0^{c_i} \ell_j(\tau)\, \mathrm{d}\tau, \qquad b_j = \int_0^1 \ell_j(\tau)\, \mathrm{d}\tau, \qquad B = \mathrm{diag}(b). \tag{88}$$

Local diagonal stage pairing is $P = B$. The compatibility equations then reduce to

$$B\mathcal{A} + \mathcal{A}^\top B = bb^\top, \tag{89}$$

the Cooper condition.

Theorem 8.5 (Symplectic collocation is exactly Gauss collocation)**.** *For a collocation method with s distinct real nodes, the following are equivalent:*

1. *the local diagonal pairing $P = B$ satisfies the chain rule;*
2. *the Cooper condition Equation (89) holds;*
3. *the interpolatory quadrature is exact on $\mathcal{P}_{2s-1}$;*
4. *the nodal polynomial $\pi(\tau) = \prod_{i=1}^s (\tau - c_i)$ is orthogonal to $\mathcal{P}_{s-1}$ in $L^2(0,1)$;*
5. *the nodes and weights are the shifted Gauss–Legendre rule.*

Proof. The first two statements are the specialization of Proposition 8.1. Put

$$L_j(t) = \int_0^t \ell_j(\tau)\, \mathrm{d}\tau.$$

The quadrature functional $\mathcal{Q}(p) = \sum_k b_k p(c_k)$ satisfies

$$\mathcal{Q}((L_iL_j)') = b_i a_{ij} + b_j a_{ji}.$$

Under the Cooper condition this equals $b_ib_j = L_i(1)L_j(1) = \int_0^1 (L_iL_j)'$. The L_i span

$$\{p \in \mathcal{P}_s : p(0) = 0\} = t\mathcal{P}_{s-1}.$$

Their pairwise products span $t^2\mathcal{P}_{2s-2}$, and their derivatives span $t\mathcal{P}_{2s-2}$, i.e. all monomials $t, t^2, \dots, t^{2s-1}$. Constants are already integrated exactly. Thus Cooper implies exactness on $\mathcal{P}_{2s-1}$. The converse follows by applying exactness to each $(L_iL_j)'$.

An s-node interpolatory rule is exact on $\mathcal{P}_{2s-1}$ exactly when, for every $q \in \mathcal{P}_{s-1}$,

$$\int_0^1 \pi(\tau) q(\tau)\, \mathrm{d}\tau = 0.$$

The unique monic degree-s polynomial with this orthogonality property is the shifted Legendre polynomial. Its roots are automatically distinct and interior to $(0,1)$; they are the Gauss nodes, and the interpolatory weights are then the Gauss weights. □

Corollary 8.6 (Polynomial energy proof)**.** *Let $u \in \mathcal{P}_s([0,1]; V)$ be a collocation polynomial for $u' = hJQu$. Then Gauss nodes imply exact conservation of $\frac{1}{2}u^\top Qu$.*

Proof. The polynomial $p(\tau) = (Qu(\tau))^\top u'(\tau)$ has degree at most $2s-1$ and vanishes at every collocation node. Hence $p = \pi q$, with $q \in \mathcal{P}_{s-1}$. Orthogonality in Theorem 8.5 gives

$$H(u(1)) - H(u(0)) = \int_0^1 p(\tau)\, \mathrm{d}\tau = 0.$$

□

Proposition 8.7 (From the local Stokes pairing to active stages)**.** *Let the nodes c_i be distinct and real, let $Q = Q^\top \neq 0$, and let $x_h \in \mathcal{P}_s([0,1]; X)$. Write its scaled flow as*

$$h^{-1}x_h'(\tau) = k_h(\tau) = \sum_{j=1}^{s} \ell_j(\tau) k_j, \qquad X_i = x_h(c_i). \tag{90}$$

Then polynomial integration gives, without invoking a Runge–Kutta ansatz,

$$X_i = x_0 + h\sum_{j=1}^{s} a_{ij}k_j, \qquad x_1 = x_0 + h\sum_{j=1}^{s} b_j k_j. \tag{91}$$

For the quadratic effort $e_h = Qx_h$, the local diagonal pairing

$$h\sum_{i=1}^{s} b_i (QX_i)^\top k_i \tag{92}$$

is the exact cap difference $H(x_1) - H(x_0)$ for every $x_h \in \mathcal{P}_s$ if and only if the Cooper condition holds, equivalently if and only if the nodes and weights are the shifted Gauss rule. If additionally $J^\top = -J$, closing the active variables by $k_i = JQX_i$ annihilates Equation (92) and turns Equation (91) into the Gauss stage equations.

Proof. Integrating Equation (90) from 0 to c_i and to 1 gives Equation (91). Moreover,

$$H(x_1) - H(x_0) = h\int_0^1 (Qx_h)^\top k_h \, \mathrm{d}\tau.$$

The integrand has degree at most $2s-1$, so Gauss exactness proves sufficiency. Conversely, suppose the pairing is exact for every x_h. Since Q is real symmetric and nonzero, some $v \in X$ satisfies $v^\top Q v \neq 0$. Taking $x_h(\tau) = p(\tau)v$ proves quadrature exactness for $pp' = \frac{1}{2}(p^2)'$, for every $p \in \mathcal{P}_s$. Polarization with $p+q$ then proves exactness for

$$(pq)' = p'q + pq' \qquad (p, q \in \mathcal{P}_s).$$

The products pq span $\mathcal{P}_{2s}$, hence their derivatives span $\mathcal{P}_{2s-1}$. Thus the quadrature is exact on the whole space $\mathcal{P}_{2s-1}$, and Theorem 8.5 gives the Cooper condition and the shifted Gauss rule. Under the constitutive law each summand is $(QX_i)^\top J(QX_i) = 0$. Thus the active stage equations follow from the local polynomial Stokes pairing plus constitutive closure, and their later Schur reduction is not an arbitrary rewriting of a preselected Runge–Kutta method. □

Theorem 8.8 (Active Gauss cell and induced cap gauge)**.** *Let $X = \mathbb{R}^n$, let $A = JQ$, and take the shifted s-point Gauss–Legendre nodes, Lagrange polynomials, matrix $\mathcal{A}_s$, and weights b from Equation (88). The polynomial temporal cell*

$$\mathcal{P}_s([0,1]; X) \xrightarrow{\partial_\tau} \mathcal{P}_{s-1}([0,1]; X)$$

is represented by active stage states X_i and stage flows k_i:

$$X_i = x_0 + h\sum_{j=1}^{s} a_{ij}k_j, \qquad k_i = AX_i, \tag{93}$$

$$x_1 = x_0 + h\sum_{j=1}^{s} b_j k_j. \tag{94}$$

Put

$$\mathcal{K}_s(h, A) = I_s \otimes I - h\mathcal{A}_s \otimes A. \tag{95}$$

Whenever this block is invertible, elimination of the sn stage variables gives

$$M_s^{\mathrm{cell}}(hA) = I + h(b^\top \otimes A)\mathcal{K}_s(h,A)^{-1}(\mathbf{1}\otimes I) = R_s(hA), \tag{96}$$

where

$$P_s(z) = \sum_{k=0}^{s} \frac{(2s-k)!\,s!}{(2s)!\,k!\,(s-k)!}\, z^k, \tag{97}$$

$$R_s(z) = P_s(-z)^{-1}P_s(z). \tag{98}$$

Assume in addition that J, Q, $R_s(hA) - I$, and the displayed matrix functions below are invertible. Write

$$E_s(z) = \tfrac{1}{2}(P_s(z) + P_s(-z)), \qquad O_s(z) = \tfrac{1}{2}(P_s(z) - P_s(-z)) = z\widehat{O}_s(z).$$

The cap gauge induced by this interior Schur reduction is

$$\boxed{S_{s,h}^{\mathrm{cell}} = \frac{1}{h}\Omega\left[\frac{1}{2}E_s(hA)\widehat{O}_s(hA)^{-1} - I\right], \qquad \Omega = J^{-1}.} \tag{99}$$

It is alternating, and its two-point map is exactly the s-stage Gauss method.

Proof. The polynomial reconstruction and the two stage relations follow from Proposition 8.7; conversely they reconstruct a unique $x_h \in \mathcal{P}_s$. Stacking the stage equations gives

$$\mathcal{K}_s(h,A)X = (\mathbf{1}\otimes I)x_0,$$

and substitution into the endpoint equation proves the first equality in Equation (96).

The scalar stability function of Gauss collocation has numerator and denominator of degree at most s and agrees with e^z through degree $2s$; it is therefore the diagonal Padé approximant $P_s(-z)^{-1}P_s(z)$. The same rational identity holds for matrices on the common resolvent set, proving the second equality.

It remains to identify the cap two-form generated by the stages. Since all polynomials in A commute,

$$(R_s + I)(R_s - I)^{-1} = E_s(hA)O_s(hA)^{-1} = (hA)^{-1}E_s(hA)\widehat{O}_s(hA)^{-1}.$$

Substitution into the inverse Cayley formula Equation (71) gives Equation (99). Alternation follows either from Theorem 5.3 and $R_s^\top Q R_s = Q$, or directly from $A^\top\Omega + \Omega A = 0$. □

Remark 8.9 (Representation versus interior generation)**.** *The trace-generated theorem Theorem 4.14 starts from S_h and realizes it in a redundant raw cell. By contrast, Theorem 8.8 starts from the standard local Gauss pairing on $\mathcal{P}_s \to \mathcal{P}_{s-1}$ and the constitutive stage law $k_i = AX_i$; the cap gauge is computed only after the active stage block has been eliminated. This is a genuine interior generation result.*

Remark 8.10. *The same monic-polynomial orthogonality is the scalar core of the spatial $\mathcal{P}_N/\mathcal{P}_{N-1}$ Stokes-pairing construction [20]. The common mechanism is not merely that both methods use Gauss nodes: in both variables, scalar locality turns compatibility into maximal-degree quadrature exactness, and maximal-degree exactness is equivalent to orthogonality of the nodal polynomial.*

9 Space–time Stokes–Dirac closure and local rigidity

9.1 Maximality, constitutive closure, and glued faces

Let active state flows and efforts lie in dual spaces of dimension d, and let a boundary port have dimension r. Coordinates may be chosen after restricting to the active quotient in Equation (33). Consider

$$\mathcal{D}_h = \left\{(f,u;e,y) : f = -\mathcal{J}_h e - Bu, \quad y = B^\top e + Nu\right\}, \qquad \mathcal{J}_h^\top = -\mathcal{J}_h, \quad N^\top = -N. \tag{100}$$

The symmetric split-power pairing is

$$\langle\langle (f,u;e,y),(f',u';e',y')\rangle\rangle_+ = e^\top f' + e'^\top f + y^\top u' + y'^\top u. \tag{101}$$

Proposition 9.1 (Finite-dimensional open Dirac structure with feedthrough)**.** *The subspace $\mathcal{D}_h$ in Equation (100) is maximally isotropic for Equation (101).*

Proof. For any element of $\mathcal{D}_h$,

$$e^\top f + y^\top u = -e^\top \mathcal{J}_h e - e^\top Bu + e^\top Bu + u^\top Nu = 0.$$

Polarization proves isotropy. The free variables (e,u) give $\dim \mathcal{D}_h = d + r$, half the dimension of the ambient split space. Therefore the isotropic subspace is maximal. □

Theorem 9.2 (Rectangular Stokes-to-Dirac reduction)**.** *Let E be a raw effort space, X an active state space, and U a port-input space. Consider linear maps*

$$P : E \to X^*, \qquad K : E \to X, \qquad \Gamma : E \to U, \qquad Y : E \to U^*.$$

Assume the rectangular Stokes identity

$$\langle Pe, Ke\rangle = \langle Ye, \Gamma e\rangle \qquad (e \in E) \tag{102}$$

and suppose that

$$\mathcal{R} = \begin{bmatrix} P \\ \Gamma \end{bmatrix} : E \longrightarrow X^* \oplus U \tag{103}$$

is an isomorphism. Define the square blocks by

$$\begin{bmatrix} \mathcal{J} & B \end{bmatrix} = K\mathcal{R}^{-1}, \qquad \begin{bmatrix} C & N \end{bmatrix} = Y\mathcal{R}^{-1}. \tag{104}$$

Then

$$\mathcal{J}^* = -\mathcal{J}, \qquad C = B^*, \qquad N^* = -N. \tag{105}$$

Consequently the raw equations reduce exactly to

$$f = \mathcal{J}\epsilon + Bu, \qquad y = B^*\epsilon + Nu, \qquad \epsilon = Pe, \quad u = \Gamma e, \quad f = Ke, \quad y = Ye, \tag{106}$$

and the oriented tuple $(-f,u;\epsilon,y)$ belongs to the maximally isotropic relation Equation (100) with $\mathcal{J}_h = \mathcal{J}$.

Proof. For arbitrary (ϵ,u), write $e = \mathcal{R}^{-1}(\epsilon,u)$. Then Equation (102) becomes

$$\langle \epsilon, \mathcal{J}\epsilon + Bu\rangle = \langle C\epsilon + Nu, u\rangle .$$

The pure quadratic terms imply $\mathcal{J}^* = -\mathcal{J}$ and $N^* = -N$; polarization of the mixed term gives $C = B^*$. Equation Equation (106) follows from the definitions, and Proposition 9.1 gives maximal isotropy. □

Theorem 9.3 (Cap-to-mantle power balance)**.** *Let H_h be a discrete Hamiltonian on a temporal cap and let $\bar{e}(x_0, x_1)$ be any compatible temporal boundary reduction of a space–time pairing, so that*

$$\bar{e}^\top \Delta x = H_h(x_1) - H_h(x_0).$$

Impose

$$\left(-\frac{\Delta x}{h}, \bar{u}; \bar{e}, \bar{y}\right) \in \mathcal{D}_h. \tag{107}$$

Then

$$H_h(x_1) - H_h(x_0) = h\, \bar{y}^\top \bar{u}. \tag{108}$$

The result holds for every active space–time gauge whose pullback gives the chosen temporal effort. The relation $\mathcal{D}_h$ supplies maximal isotropy, whereas the compatible pairing supplies constitutive closure; neither statement implies the other.

Proof. Self-isotropy of Equation (107) gives $-\bar{e}^\top \Delta x/h + \bar{y}^\top \bar{u} = 0$. The chain rule converts the cap term into the Hamiltonian increment. Gauge independence follows because an alternating two-form vanishes when evaluated twice on the same active increment. □

Corollary 9.4 (Dissipation and gluing)**.** *If a resistive relation adds $R_h = R_h^\top \succeq 0$, then*

$$H_h(x_1) - H_h(x_0) = h\, \bar{y}^\top \bar{u} - h\, \bar{e}^\top R_h \bar{e}. \tag{109}$$

If two space–time cells share a face and are interconnected with opposite orientations, $\bar{u}_1 = -\bar{y}_2$ and $\bar{u}_2 = \bar{y}_1$, their internal supplies cancel. Summing Equation (109) over a mesh leaves only the external mantle and the resistive volume terms, exactly as internal faces cancel in a cellular boundary.

9.2 Local diagonal rigidity derived from the total equations

Let c_i^t, $1 \le i \le N_t$, and c_j^x, $1 \le j \le N_x$, be distinct nodes with Lagrange bases ℓ_i^t and ℓ_j^x. A pointwise diagonal pairing on the tensor grid is determined by a weight array $W = (W_{ji})$ through

$$\langle p, q \rangle_W = \sum_{i=1}^{N_t} \sum_{j=1}^{N_x} W_{ji}\, p(c_i^t, c_j^x) q(c_i^t, c_j^x). \tag{110}$$

Suppose the one-dimensional local compatible weight spaces are the lines $\mathrm{span}\{w^t\}$ and $\mathrm{span}\{w^x\}$. This conclusion follows from Theorem 8.5 for maximal polynomial collocation and from its spatial Stokes analogue.

Theorem 9.5 (Tensor rigidity from global Stokes compatibility)**.** *Assume that the full tensor trial space contains all products $\ell_i^t p_x$ and $p_t \ell_j^x$, and that the diagonal pairing Equation (110) satisfies the two components of the global Stokes identity associated with D_{st}. Then*

$$W = \alpha\, w^x (w^t)^\top \tag{111}$$

for a scalar α. If the one-dimensional weights and W are normalized, $\alpha = 1$. If the factor rules are polynomial rules of maximal degree, the unique normalized local pairing is

$$\boxed{W = w^x (w^t)^\top = \mathrm{GL}_x \otimes \mathrm{GL}_t.} \tag{112}$$

Proof. Fix a temporal node c_i^t and test the spatial component of the global identity with functions $\ell_i^t p_x$ and $\ell_i^t q_x$. Since $\ell_i^t(c_k^t) = \delta_{ik}$, every temporal slice except i vanishes in Equation (110). The column $W_{:i}$ must therefore be a one-dimensional spatial compatible weight vector. Hence

$$W_{:i} = \lambda_i w^x, \qquad W = w^x \lambda^\top.$$

Now fix a spatial node and use $p_t\ell_j^x$, $q_t\ell_j^x$ in the temporal component. Every row of W belongs to $\mathrm{span}\{(w^t)^\top\}$. Choosing j with $w_j^x \neq 0$ gives $\lambda = \alpha w^t$, proving Equation (111). Normalization fixes $\alpha = 1$, and maximal polynomial exactness selects the Gauss factors by the nodal-polynomial argument in Theorem 8.5, applied on each axis. □

Remark 9.6 (What locality removes)**.** *The conclusion is not a uniqueness statement in the full affine family Equation (32). The local diagonal ansatz removes the nonlocal sectors in Equation (37); the global Stokes equations then force separability. A nonzero mixed gauge is therefore a precise algebraic obstruction to representing the pairing by pointwise product weights, not a violation of Stokes compatibility.*

9.3 A wave system with boundary ports

Consider the first-order wave system on a spatial nodal grid. Let $W_x = W_x^\top \succ 0$ and D_x satisfy the summation-by-parts identity

$$W_x D_x + D_x^\top W_x = B_x, \qquad B_x = \mathrm{diag}(-1, 0, \dots, 0, 1). \tag{113}$$

For $x = (q, p)$, set

$$Q_h = \mathrm{diag}(W_x, W_x), \qquad \mathcal{J}_h = \begin{bmatrix} 0 & D_x W_x^{-1} \\ D_x W_x^{-1} & 0 \end{bmatrix}, \qquad \mathcal{R}_h = \rho Q_h^{-1}, \quad \rho \geq 0. \tag{114}$$

Here $\mathcal{J}_h$ is the unclosed Stokes operator, not a skew matrix: its symmetric defect is exactly the boundary term encoded by Equation (113). Introducing the two endpoint ports converts this defect into an open Dirac relation of the form Equation (100). For any skew gauge S, define

$$g = Q_h \bar{x} - S\Delta x, \qquad v = Q_h^{-1} g = (v_q, v_p), \tag{115}$$

and close the cell by

$$\frac{\Delta x}{h} = (\mathcal{J}_h - \mathcal{R}_h) g. \tag{116}$$

Proposition 9.7 (Gauge-independent wave power)**.** *Every well-defined solution of Equation (116) satisfies*

$$\begin{aligned} H_h(x_1) - H_h(x_0) = {} & h(v_{q,N} v_{p,N} - v_{q,0} v_{p,0}) \\ & - h\rho(v_q^\top W_x v_q + v_p^\top W_x v_p), \end{aligned} \tag{117}$$

where $H_h(x) = \frac{1}{2} x^\top Q_h x$. The first line is the right-face power minus the left-face power. On two adjacent cells, continuity of the two boundary efforts makes the shared-face contributions cancel with opposite orientations.

Proof. Skewness of S gives

$$H_h(x_1) - H_h(x_0) = g^\top \Delta x.$$

Moreover, Equation (113) yields

$$g^\top \mathcal{J}_h g = v_q^\top W_x D_x v_p + v_p^\top W_x D_x v_q = v_q^\top B_x v_p,$$

which is the oriented boundary term in Equation (117); and $g^\top \mathcal{R}_h g = \rho(v_q^\top W_x v_q + v_p^\top W_x v_p)$. Substitute Equation (116). Shared faces occur twice with opposite entries of B_x, proving cancellation. □

9.4 A direct rectangular transmission-line cylinder

We now apply the cylinder gauge to the polynomial reduction of the ideal transmission line in [20], rather than first selecting a spatial state-space model and then applying an unrelated time integrator. Let

$$E_N = \mathcal{P}_N([0,\ell]), \qquad F_N = \mathcal{P}_{N-1}([0,\ell]), \qquad D_N = \partial_z : E_N \longrightarrow F_N,$$

and let the full-rank rectangular pairing $M_N : E_N \times F_N \to \mathbb{R}$ satisfy

$$M_N D_N + D_N^\top M_N^\top = r_\ell r_\ell^\top - r_0 r_0^\top, \tag{118}$$

where $r_0, r_\ell \in E_N^*$ are endpoint evaluation. This is the coordinate form of the exact polynomial Stokes identity used in the 2012 construction. No square same-grid differentiation matrix is assumed. The temporal factor is the minimal exact complex $\mathcal{P}_1([0,h]) \xrightarrow{\partial_t} \mathcal{P}_0([0,h])$: its traces are the two caps and its single flow coordinate is the cap difference divided by h. Consequently the following equations are the simultaneous closure of one tensor cylinder cell, even though they may subsequently be eliminated to a cap map.

On a temporal slab, take cap states $x_a = (q_a, p_a) \in F_N \oplus F_N$, $a \in \{0,1\}$, and spatial effort polynomials $e_q, e_p \in E_N$. Put

$$\epsilon = \begin{bmatrix} M_N^\top e_q \\ M_N^\top e_p \end{bmatrix}, \qquad H_N(x) = \tfrac{1}{2} x^\top Q_N x, \qquad Q_N = Q_N^\top \succ 0. \tag{119}$$

For a skew cap gauge $S_h^\top = -S_h$, impose simultaneously

$$\frac{q_1 - q_0}{h} = -D_N e_p, \qquad \frac{p_1 - p_0}{h} = -D_N e_q, \tag{120}$$

$$\epsilon = Q_N \frac{x_0 + x_1}{2} - S_h(x_1 - x_0). \tag{121}$$

Choose the boundary causality

$$u = \begin{bmatrix} e_q(0) \\ e_p(\ell) \end{bmatrix}, \qquad y = \begin{bmatrix} e_p(0) \\ -e_q(\ell) \end{bmatrix}. \tag{122}$$

We denote these oriented components by $u = (u_L, u_R)^\top$ and $y = (y_L, y_R)^\top$, so that the left supply is $y_L u_L$ and the right supply is $y_R u_R$. This convention is used unchanged when neighboring cells are interconnected. The homogeneous conditions $u = 0$ are the complementary endpoint conditions used for the closed ideal line in [20].

Proposition 9.8 (Direct cap–mantle balance for the rectangular line)**.** *Every well-defined solution of Equations (120) and (121) satisfies*

$$H_N(x_1) - H_N(x_0) = h\, y^\top u. \tag{123}$$

If two such cells are glued by $u_{1,R} = y_{2,L}$ and $u_{2,L} = -y_{1,R}$, their interface supplies cancel. Adding a symmetric positive semidefinite active resistance R_N to the flow relation changes the right-hand side to

$$h\, y^\top u - h\, \epsilon^\top R_N \epsilon.$$

Proof. Skewness of S_h and Equation (121) give

$$\begin{aligned} H_N(x_1) - H_N(x_0) &= \epsilon_q^\top (q_1 - q_0) + \epsilon_p^\top (p_1 - p_0) \\ &= -h(e_q^\top M_N D_N e_p + e_p^\top M_N D_N e_q) \end{aligned}$$

$$= -h\, e_q^\top (M_N D_N + D_N^\top M_N^\top) e_p.$$

Using Equation (118), the last expression is

$$h(e_q(0)e_p(0) - e_q(\ell)e_p(\ell)) = h\, y^\top u.$$

The interconnection contribution is $y_{1,R}y_{2,L} - y_{2,L}y_{1,R} = 0$. The resistive statement follows by contracting the additional flow $-R_N\epsilon$ with ϵ. □

The cap gauge in Equation (121) can now be placed inside the exact visibility sequence of Theorem 4.10. Write $\mathcal{D}_x = I_2 \otimes D_N$ and define the cap trace

$$\mathcal{T}_N(z_0, z_1) = \mathcal{D}_x(z_1 - z_0), \qquad \mathcal{T}_N : (E_N \oplus E_N)^2 \to F_N \oplus F_N. \tag{124}$$

For an alternating form $\Theta \in \Lambda^2(F_N \oplus F_N)^*$, define

$$\omega_{tx}^\Theta = \mathcal{T}_N^*\Theta, \qquad \omega_{tx}^\Theta(z, w) = \Theta(\mathcal{T}_N z, \mathcal{T}_N w). \tag{125}$$

Proposition 9.9 (Surjective mixed realization of all cap gauges)**.** *The form ω_{tx}^Θ is well defined on the active space–time quotient. It vanishes on time-constant fields and on fields whose spatial part lies in $\ker \mathcal{D}_x$; hence it is supported on the mixed sector Q_{tx} of Equation (36). Equivalently, after passage to the active quotient, there is a unique onto map $\widetilde{\mathcal{T}}_N$ such that*

$$\boxed{\mathcal{T}_N = \widetilde{\mathcal{T}}_N \pi_{tx}.} \tag{126}$$

Moreover, $\mathcal{T}_N$ is onto. If $\sigma_x : F_N \oplus F_N \to E_N \oplus E_N$ is any right inverse of $\mathcal{D}_x$, then

$$\mathcal{L}_N \xi = (0, \sigma_x \xi) \tag{127}$$

is an explicit geometric right inverse with $\mathcal{T}_N\mathcal{L}_N = I$. More generally, every regular interior reconstruction L_h consistent with the cap trace satisfies $\mathcal{T}_N L_h = I$, and

$$\boxed{\mathcal{R}_{L_h}(\omega_{tx}^\Theta) = L_h^*\mathcal{T}_N^*\Theta = \Theta.} \tag{128}$$

Consequently every alternating cap gauge on the active line state has an explicit purely mixed cylinder realization.

Proof. Changing either cap potential by an element of $\ker \mathcal{D}_x$ leaves Equation (125) unchanged. If $z_1 = z_0$, the temporal increment vanishes; if both spatial increments lie in the spatial kernel, their images under $\mathcal{D}_x$ vanish. Alternation follows from $\Theta^\top = -\Theta$. These are precisely the quotient and support properties defining the Q_{tx} sector. The universal property of the quotient gives Equation (126). Since $D_N : \mathcal{P}_N \to \mathcal{P}_{N-1}$ is onto, $\mathcal{D}_x$ and $\mathcal{T}_N$ are onto and a right inverse σ_x exists. Finally,

$$L_h^*\mathcal{T}_N^*\Theta = (\mathcal{T}_N L_h)^*\Theta = \Theta.$$

This is the injective-reconstruction case of Theorem 4.10. □

9.5 Interior Schur generation versus encoded realization

The pullback in Proposition 9.9 places a prescribed two-form in the cylinder. A different question is whether eliminating genuine auxiliary variables can produce the same cap form. The following finite-dimensional result gives a sharp algebraic answer.

Theorem 9.10 (Minimal auxiliary Schur generation)**.** *Let V be an n-dimensional cap space and let $\Theta_0, \Theta_1 \in \Lambda^2 V^*$. Put $\Delta = \Theta_1 - \Theta_0$ and $\operatorname{rank}\Delta = 2q$. Then there exist an interior space W of dimension $2q$, a nondegenerate alternating matrix $K : W \to W^*$, and a coupling $B : W \to V^*$ such that*

$$\mathbb{G} = \begin{bmatrix} \Theta_0 & B \\ -B^\top & K \end{bmatrix}, \qquad \mathbb{G}^\top = -\mathbb{G}, \tag{129}$$

and elimination of the interior variable from

$$p = \Theta_0 x + B\eta, \qquad 0 = -B^\top x + K\eta \tag{130}$$

gives exactly

$$p = \Theta_1 x. \tag{131}$$

Moreover, $2q$ is the smallest possible interior dimension among all representations of the form Equation (129) with invertible K.

Proof. The real congruence normal form for an alternating matrix gives a full column-rank matrix $U \in \mathbb{R}^{n\times 2q}$ such that

$$\Delta = U\mathbb{J}_q U^\top, \qquad \mathbb{J}_q = \operatorname{diag}\left(\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \dots, \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}\right).$$

Take $B = U$ and $K = -\mathbb{J}_q$, so that $K^{-1} = \mathbb{J}_q$. The second equation in Equation (130) gives $\eta = K^{-1}B^\top x$, and substitution in the first gives

$$p = (\Theta_0 + BK^{-1}B^\top)x = (\Theta_0 + \Delta)x = \Theta_1 x.$$

The displayed block matrix is alternating by construction. Conversely, every such elimination satisfies

$$\Delta = BK^{-1}B^\top, \qquad \operatorname{rank}\Delta \le \dim W.$$

Thus no smaller interior space can generate a rank-$2q$ change. □

Remark 9.11 (What the theorem does not select)**.** *Theorem 9.10 is stronger than a volume pullback: the target gauge appears only after an interior variable is eliminated, and the number of auxiliary modes is minimal. It is still an* encoded auxiliary generation*, because B and K are built from the desired Δ. It does not prove that a spatial–temporal field complex, specified independently of Θ_1, selects that gauge. That field-selection problem remains the genuinely mixed open step. The theorem's role is therefore to close the unrestricted algebraic existence question and identify its sharp dimension. Its block factorization is an application of the standard real normal form for an alternating matrix, not a claim of a new general realization theory.*

The rectangular maps entering Theorem 9.2 are

$$\mathcal{M}_N^* = \operatorname{diag}(M_N^\top, M_N^\top), \qquad \mathcal{K}_N = \begin{bmatrix} 0 & -D_N \\ -D_N & 0 \end{bmatrix}, \tag{132}$$

while $\mathcal{C}_N e = u$ and $\mathcal{Y}_N e = y$ are the input and output traces in Equation (122).

Lemma 9.12 (Polynomial endpoint reconstruction)**.** *For either $a = 0$ or $a = \ell$, the single-field map*

$$\begin{bmatrix} M_N^\top \\ r_a^\top \end{bmatrix} : E_N \longrightarrow F_N^* \oplus \mathbb{R} \tag{133}$$

is an isomorphism. Consequently, with the complementary causality in Equation (122), the two-field effort/input reconstruction $\mathcal{R}_N$ below is an isomorphism.

Proof. Suppose first that $M_N^\top e = 0$ and $r_0^\top e = 0$. Contracting Equation (118) on both sides with e gives $(r_\ell^\top e)^2 - (r_0^\top e)^2 = 0$, hence $r_\ell^\top e = 0$. Left multiplication of the same identity by $e^\top$ now gives

$$(D_N e)^\top M_N^\top = 0,$$

because both trace terms and $e^\top M_N$ vanish. Thus $M_N D_N e = 0$. The pairing matrix M_N has full column rank, so $D_N e = 0$. Therefore e is constant, and its zero endpoint trace forces $e = 0$. The proof with $r_\ell^\top e = 0$ is identical, with the endpoints exchanged. Each map in Equation (133) is injective between spaces of the same dimension $N + 1$, hence bijective. Taking the block sum of the r_0 reconstruction for e_q and the r_ℓ reconstruction for e_p proves the last claim. □

The map

$$\mathcal{R}_N = \begin{bmatrix} \mathcal{M}_N^* \\ \mathcal{C}_N \end{bmatrix} \tag{134}$$

is therefore invertible: each full effort polynomial is determined by its N active power coordinates and its prescribed complementary endpoint trace. The rectangular identity Equation (118) is exactly Equation (102). Therefore Theorem 9.2 proves, rather than assumes, the finite-dimensional representation

$$f = \mathcal{J}_N \epsilon + B_N u, \qquad y = C_N \epsilon + N_N u, \qquad C_N = B_N^\top, \quad \mathcal{J}_N^\top = -\mathcal{J}_N, \quad N_N^\top = -N_N. \tag{135}$$

The blocks are explicitly

$$\begin{bmatrix} \mathcal{J}_N & B_N \end{bmatrix} = \mathcal{K}_N \mathcal{R}_N^{-1}, \qquad \begin{bmatrix} B_N^\top & N_N \end{bmatrix} = \mathcal{Y}_N \mathcal{R}_N^{-1}. \tag{136}$$

This square representation is a consequence of the rectangular Stokes identity; it is not the starting point. For the closed line put $A_N = \mathcal{J}_N Q_N$. If $f_{\mathrm{phys}} = \Delta x/h$, the correct oriented Dirac statement is

$$(-f_{\mathrm{phys}}, u; \epsilon, y) \in \mathcal{D}_h, \qquad \mathcal{J}_h = \mathcal{J}_N, \quad B = B_N, \quad N = N_N, \tag{137}$$

in agreement with Equation (107).

Theorem 9.13 (Mixed deformations of the stage-induced Gauss hierarchy)**.** *Assume that Q_N and $\mathcal{J}_N$ are nonsingular. Let $\Theta^\top = -\Theta$ be the cap reduction of Equation (125). For $s \geq 1$, let $P_s, E_s, \widehat{O}_s$, and R_s be those of Theorem 8.8. Where the displayed inverses exist, define the stage-induced Gauss–s cap gauge by*

$$S_{s,h}^{\mathrm{G}} = \frac{1}{h}\Omega_N \left[\frac{1}{2} E_s(hA_N)\widehat{O}_s(hA_N)^{-1} - I\right], \qquad \Omega_N = \mathcal{J}_N^{-1}. \tag{138}$$

Define its mixed deformation by

$$\boxed{S_{s,h}^{\Theta} = S_{s,h}^{\mathrm{G}} + h^{2s-1}\Theta.} \tag{139}$$

For $u = 0$, the step

$$\Delta x = h\mathcal{J}_N \left(Q_N \bar{x} - S_{s,h}^{\Theta} \Delta x\right) \tag{140}$$

has the following properties whenever its matrices are invertible.

1. *At $\Theta = 0$, $S_{s,h}^{\mathrm{G}}$ is generated by elimination of the s active polynomial stages in Theorem 8.8. The added mixed perturbation has the volume representative $\mathcal{T}_N^*(h^{2s-1}\Theta)\mathcal{T}_N$, and exact raw-cell elimination recovers the sum $S_{s,h}^{\Theta}$.*

2. *It preserves H_N exactly, is self-adjoint, and has global order at least $2s$. At $\Theta = 0$, its propagator is the diagonal Padé map $R_s(hA_N)$, equivalently the s-stage Gauss method.*

3. *It is symplectic if and only if*
$$[\mathcal{J}_N\Theta, A_N] = 0. \tag{141}$$

4. *If $[\mathcal{J}_N\Theta, A_N] \neq 0$, then for all sufficiently small nonzero h its propagator does not commute with A_N. It therefore lies outside the scalar functional calculus of A_N. In particular, it cannot be the result of applying any Runge–Kutta or collocation method with scalar coefficients to $\dot{x} = A_N x$.*

Proof. Theorem 8.8, applied with $A = A_N$ and $\Omega = \Omega_N$, proves that $S^{\mathrm{G}}_{s,h}$ is alternating and generated by the interior Gauss cell, with cap map $R_s(hA_N)$. The mixed lift follows from Proposition 9.9; its exact raw realization is Theorem 4.14.

Put
$$B_{s,\pm} = I + h\mathcal{J}_N S^{\mathrm{G}}_{s,h} \pm \frac{h}{2}A_N, \qquad E_{s,h} = h^{2s}\mathcal{J}_N\Theta.$$
Eliminating the effort from Equation (140) gives
$$M_{s,\Theta,h} = (B_{s,-} + E_{s,h})^{-1}(B_{s,+} + E_{s,h}), \qquad B_{s,-}^{-1}B_{s,+} = R_s(hA_N). \tag{142}$$
The discrete chain rule proves exact energy preservation. Both $S^{\mathrm{G}}_{s,h}$ and $h^{2s-1}\Theta$ are odd in h, so reversing h and exchanging the caps inverts the map; hence the method is self-adjoint.

The exact resolvent identity
$$M_{s,\Theta,h} - R_s = (B_{s,-} + E_{s,h})^{-1}E_{s,h}(I - R_s)$$
is $O(h^{2s+1})$. Since diagonal Padé has local error $O(h^{2s+1})$, the deformation has local error $O(h^{2s+1})$ and therefore global order at least $2s$. The leading coefficient and the unique exceptional value of Θ are computed exactly in Theorem 9.14 below.

The symplectic criterion Equation (76), applied at each fixed h, is automatic for $S^{\mathrm{G}}_{s,h}$, because its map is Gauss and $\mathcal{J}_N S^{\mathrm{G}}_{s,h}$ is a rational function of A_N. The remaining condition is $\Theta A_N = Q_N\mathcal{J}_N\Theta$; multiplication by $\mathcal{J}_N$ gives Equation (141). Finally, $E_{s,h} = h^{2s}\mathcal{J}_N\Theta$ and $I - R_s = -hA_N + O(h^2)$ yield
$$[M_{s,\Theta,h}, A_N] = -h^{2s+1}[\mathcal{J}_N\Theta, A_N]A_N + O(h^{2s+2}). \tag{143}$$
Nonsingularity of A_N proves the noncommutation claim. Every scalar rational, polynomial, or analytic function of A_N commutes with it. □

Theorem 9.14 (Exact one-term order dichotomy)**.** *Under the assumptions of Theorem 9.13, set*
$$c_s = (-1)^{s+1}\frac{(s!)^2}{(2s)!(2s+1)!}. \tag{144}$$
Within the one-term ansatz Equation (139), let Θ be alternating, independent of h, and fixed as $h \to 0$. Then
$$M_{s,\Theta,h} - e^{hA_N} = h^{2s+1}(c_s A_N^{2s} - \mathcal{J}_N\Theta)A_N + O(h^{2s+2}). \tag{145}$$
Consequently the unique h-independent leading coefficient in this ansatz that cancels the leading error is
$$\boxed{\Theta_s^\star = c_s\Omega_N A_N^{2s}.} \tag{146}$$

It is alternating. If $\Theta \neq \Theta_s^\star$, the method has exact global order $2s$; for $\Theta = \Theta_s^\star$, it has global order exactly $2s+2$. More precisely,

$$M_{s,\Theta_s^\star,h} - e^{hA_N} = d_s h^{2s+3} A_N^{2s+3} + O(h^{2s+4}), \tag{147}$$

where

$$d_1 = -\frac{1}{720}, \qquad d_s = -c_s \frac{(s+1)(2s-3)}{6(2s-1)(2s+3)} \quad (s \geq 2). \tag{148}$$

In particular, every noncommuting coefficient in the one-term ansatz has exactly order $2s$. The cap coefficient $\Theta_s^\star$ depends on the constitutive generator, but by Proposition 9.9 it possesses a pure-mixed cylinder realization. For $s = 1$, the order-raised method is identically the two-stage Gauss method, not merely a fourth-order approximation to it.

Proof. Write $P_s(z) = \sum_{k=0}^{s} a_k z^k$, with a_k given in Equation (97). For $r \geq 0$, coefficient extraction and the beta integral give

$$\begin{aligned} g_{s,r} &:= [z^{2s+1+r}](P_s(-z)e^z - P_s(z)) \\ &= \frac{s!}{(2s)!r!} \int_0^1 (1-t)^r \sum_{k=0}^{s} \frac{(-1)^k t^{2s-k}}{k!(s-k)!}\, \mathrm{d}t \\ &= (-1)^s \frac{s!(s+r)!}{(2s)!r!(2s+r+1)!}. \end{aligned} \tag{149}$$

Indeed,

$$\sum_{k=0}^{s} \frac{(-1)^k t^{2s-k}}{k!(s-k)!} = \frac{(-1)^s}{s!} t^s (1-t)^s,$$

which proves Equation (149). Since

$$a_1 = \frac{1}{2}, \qquad a_2 = \frac{s-1}{4(2s-1)},$$

division by $P_s(-z) = 1 - a_1 z + a_2 z^2 + O(z^3)$, using $r = 0, 1, 2$, yields

$$R_s(z) - e^z = c_s z^{2s+1}(1 + z + \rho_s z^2 + O(z^3)), \qquad \rho_s = \frac{8s^2 + 10s - 5}{4(2s-1)(2s+3)}. \tag{150}$$

In particular, its first term is

$$R_s(z) - e^z = c_s z^{2s+1} + O(z^{2s+2}). \tag{151}$$

For the mixed perturbation, the exact resolvent identity in the proof of Theorem 9.13, together with $E_{s,h} = h^{2s} \mathcal{J}_N \Theta$, gives

$$M_{s,\Theta,h} - R_s(hA_N) = -h^{2s+1} \mathcal{J}_N \Theta A_N + O(h^{2s+2}).$$

Adding Equation (151) proves Equation (145). Since A_N and $\mathcal{J}_N$ are invertible, its coefficient vanishes if and only if $\Theta = c_s \Omega_N A_N^{2s}$. Moreover,

$$(\Omega_N A_N^{2s})^\top = -\Omega_N A_N^{2s},$$

so this unique coefficient is an admissible two-form. Every other choice has nonzero local error at degree $2s+1$, hence exact global order $2s$. Cancellation gives global order at least $2s+1$; self-adjoint one-step methods have even order, so the order is at least $2s+2$. If the gauge is noncommuting, it cannot equal $\Theta_s^\star$, because $\mathcal{J}_N \Theta_s^\star = c_s A_N^{2s}$ commutes with A_N. Finally, $S_{1,h}^{\mathrm{G}} = 0$, while $h\Theta_1^\star = (h/12)\Omega_N A_N^2 = S_{2,h}^{\mathrm{G}}$; the two maps are therefore identical.

It remains to determine the exact order on the raised branch. Introduce the even common blocks

$$C_s(z) = \tfrac{1}{2}E_s(z)\widehat{O}_s(z)^{-1}, \qquad C_{\mathrm{ex}}(z) = \frac{z}{2}\coth\!\left(\frac{z}{2}\right).$$

The scalar identity

$$C_{\mathrm{ex}}(z) - C_s(z) = \frac{z(R_s(z) - e^z)}{(e^z - 1)(R_s(z) - 1)} \tag{152}$$

transfers the Padé residual to the gauge coordinate. For $s \geq 2$, Equation (150) and

$$\frac{z^2}{(e^z - 1)(R_s(z) - 1)} = \left(\frac{z}{e^z - 1}\right)^2 + O(z^{2s}) = 1 - z + \frac{5}{12}z^2 + O(z^3)$$

give

$$C_{\mathrm{ex}}(z) - C_s(z) = c_s z^{2s} + d_s z^{2s+2} + O(z^{2s+4}), \tag{153}$$

where

$$d_s = c_s\left(\rho_s - \frac{7}{12}\right) = -c_s\frac{(s+1)(2s-3)}{6(2s-1)(2s+3)}.$$

For $s = 1$, $C_1 = 1$, so the Bernoulli expansion of C_{ex} gives $d_1 = -1/720$ directly. Thus the corrected common block $C_s + c_s z^{2s}$ differs from C_{ex} by $-d_s z^{2s+2} + O(z^{2s+4})$. The same resolvent identity used above, now with $z = hA_N$, proves Equation (147). Every d_s is nonzero, and A_N is invertible, so the local order is exactly $2s+3$ and the global order exactly $2s+2$. □

Corollary 9.15 (Order, centrality, and nonseparability at one term)**.** *Within Equation (139), the following are equivalent:*

1. *the global order is greater than* $2s$*;*
2. $\Theta = \Theta_s^\star$*;*
3. *the global order is exactly* $2s + 2$*.*

This unique higher-order branch lies in the symplectic centralizer and is a scalar rational function of A_N*. Conversely, every coefficient with* $[\mathcal{J}_N\Theta, A_N] \neq 0$ *has exactly order* $2s$*, is nonsymplectic for small nonzero* h*, and lies outside the scalar functional calculus of* A_N*.*

Proof. The order equivalences follow from Theorem 9.14. Moreover, $\mathcal{J}_N\Theta_s^\star = c_s A_N^{2s}$, so the exceptional coefficient commutes with A_N and satisfies Equation (141). The remaining statements follow from Equation (143) and Theorem 9.13. □

Theorem 9.16 (Two-term high-order noncommuting family)**.** *Let* $\Xi^\top = -\Xi$ *be independent of* h*, and set*

$$S_{s,h}^{\star,\Xi} = S_{s,h}^{\mathrm{G}} + h^{2s-1}\Theta_s^\star + h^{2s+1}\Xi. \tag{154}$$

The associated method is energy preserving, self-adjoint, and has global order at least $2s + 2$*. Its leading error is*

$$M_{s,\star,\Xi,h} - e^{hA_N} = h^{2s+3}(d_s A_N^{2s+2} - \mathcal{J}_N\Xi)A_N + O(h^{2s+4}). \tag{155}$$

Hence it has exact order $2s + 2$ *unless*

$$\Xi = \Xi_s^\star := d_s\Omega_N A_N^{2s+2}, \tag{156}$$

in which case its order is at least $2s + 4$*. If* $[\mathcal{J}_N\Xi, A_N] \neq 0$*, the method has exact order* $2s + 2$ *while being nonsymplectic and outside the scalar functional calculus of* A_N *for all sufficiently small nonzero* h*.*

Proof. The first correction gives Equation (147). The additional cap term changes the common block by $h^{2s+2}\mathcal{J}_N\Xi$. The resolvent identity therefore contributes $-h^{2s+3}\mathcal{J}_N\Xi A_N + O(h^{2s+4})$, proving Equation (155). Invertibility of A_N and $\mathcal{J}_N$ gives the unique second corrector. Cancellation and self-adjointness imply order at least $2s+4$. Finally,

$$[M_{s,\star,\Xi,h}, A_N] = -h^{2s+3}[\mathcal{J}_N\Xi, A_N]A_N + O(h^{2s+4}),$$

and the symplectic criterion reduces to the same commutator. □

Corollary 9.17 (Recursive central correction hierarchy)**.** *Near the origin, define the even Taylor coefficients by*

$$C_{\mathrm{ex}}(z) - C_s(z) = \sum_{j\geq 0} \alpha_{s,j} z^{2s+2j}, \qquad \alpha_{s,0} = c_s, \quad \alpha_{s,1} = d_s. \tag{157}$$

For $r \geq 1$, the truncated cap gauge

$$S_{s,h}^{[r]} = S_{s,h}^{\mathrm{G}} + \sum_{j=0}^{r-1} \alpha_{s,j} h^{2s-1+2j} \Omega_N A_N^{2s+2j} \tag{158}$$

is energy preserving, self-adjoint, symplectic, and has global order at least $2s+2r$. If $\alpha_{s,r} \neq 0$, that order is exact. Within the common disk of analyticity, the full convergent series is the exact-flow gauge. At level r, an arbitrary alternating addition $h^{2s-1+2r}\Xi$ preserves order at least $2s+2r$; it introduces noncommutation whenever $[\mathcal{J}_N\Xi, A_N] \neq 0$.

Proof. Multiplication by $h\mathcal{J}_N$ turns the j-th cap correction in Equation (158) into

$$\alpha_{s,j}(hA_N)^{2s+2j}.$$

Thus the common block agrees with $C_{\mathrm{ex}}(hA_N)$ through degree $2s+2r-2$, and the Cayley map has local error $O(h^{2s+2r+1})$. Every central correction is alternating and commutes with A_N, proving the structural claims. The last statement follows from the same one-order shift between a cap-gauge term and its map perturbation used in the preceding theorem. □

Map of the central notation. The construction passes from the diagonal Padé polynomial to a Cayley coordinate, adds the central Taylor correction, clears the coordinate denominator, and only then forms the intrinsic propagator:

$$P_s \;\longrightarrow\; (E_s, \widehat{O}_s) \;\longrightarrow\; C_s = \frac{E_s}{2\widehat{O}_s} \xrightarrow{+\delta_{s,r}} C_{s,r} \;\longrightarrow\; \Pi_{s,r} \;\longrightarrow\; R_{s,r} = \frac{\Pi_{s,r}(z)}{\Pi_{s,r}(-z)}.$$

Here $P_s(z)/P_s(-z)$ is the Gauss–s map, $\delta_{s,r}$ is the truncated central correction, $C_{s,r}$ is a chart coordinate and may diverge, whereas $\Pi_{s,r}$ is the denominator-cleared polynomial whose coprime quotient determines the true poles of $R_{s,r}$.

Theorem 9.18 (Central rational hierarchy)**.** *Let the even functions C_s, C_{ex}, and the coefficients $\alpha_{s,j}$ be defined by Equation (157). For $r \geq 1$, set*

$$\delta_{s,r}(z) = \sum_{j=0}^{r-1} \alpha_{s,j} z^{2s+2j}, \qquad \Pi_{s,r}(z) = P_s(z) + 2\widehat{O}_s(z)\delta_{s,r}(z). \tag{159}$$

Then the central propagator has the two equivalent representations

$$C_{s,r}(z) = C_s(z) + \delta_{s,r}(z), \qquad R_{s,r}(z) = \big(C_{s,r}(z) - z/2\big)^{-1}\big(C_{s,r}(z) + z/2\big) = \Pi_{s,r}(-z)^{-1}\Pi_{s,r}(z). \tag{160}$$

The last expression is understood after cancellation of common scalar polynomial factors. It has the following properties.

1. *$R_{s,r}(-z) = R_{s,r}(z)^{-1}$. Hence the induced one-step map is self-adjoint.*
2. *If $A_N^\top Q_N + Q_N A_N = 0$, then $R_{s,r}(hA_N)^\top Q_N R_{s,r}(hA_N) = Q_N$. Thus the map preserves the quadratic Hamiltonian exactly.*
3. *For fixed A_N,*
$$R_{s,r}(hA_N) - e^{hA_N} = O(h^{2s+2r+1}). \tag{161}$$
In the terminology of Definition 1.1, the local matrix defect is of order at least $2s+2r+1$, so the global operator order is at least $2s+2r$. It is exactly $2s+2r$ when $\alpha_{s,r} \neq 0$ and $A_N^{2s+2r+1} \neq 0$.
4. *Let $N_{s,r}/D_{s,r}$ be the coprime scalar reduction of $\Pi_{s,r}(z)/\Pi_{s,r}(-z)$. The true finite-step poles of the matrix propagator are exactly the failures of invertibility of $D_{s,r}(hA_N)$. Zeros of $\widehat{O}_s$ are singularities of the coordinate C_s, not additional poles of $R_{s,r}$: the polynomial representation extends the propagator through them, with $R_{s,r} = 1$.*
5. *The reduced scalar propagator has no pole on the imaginary axis. For every real ξ, its continuous rational extension satisfies*
$$|R_{s,r}(i\xi)| = 1, \qquad \arg R_{s,r}(i\xi) = 2\,\mathrm{atan2}(\Im\Pi_{s,r}(i\xi), \Re\Pi_{s,r}(i\xi)) \pmod{2\pi}. \tag{162}$$
In a finite $C_{s,r}$-chart this becomes
$$\arg R_{s,r}(i\xi) = 2\,\mathrm{atan2}(\xi, 2C_{s,r}(i\xi)) \pmod{2\pi}. \tag{163}$$

The cap gauge generating $R_{s,r}(hA_N)$ is exactly Equation (158).

Proof. Since
$$C_{s,r}(z) \pm \frac{z}{2} = \frac{P_s(\pm z) + 2\widehat{O}_s(z)\delta_{s,r}(z)}{2\widehat{O}_s(z)},$$
the coordinate denominator cancels and gives the polynomial quotient in Equation (160). This proves in particular that poles of C_s are not automatically poles of the propagator. At a zero of $\widehat{O}_s$, the coprimality of the diagonal Padé pair gives $P_s(z) = P_s(-z) = E_s(z) \neq 0$; hence the continued value is 1.

The function $C_{s,r}$ is even, which gives the reversibility identity. From $A_N^\top Q_N = -Q_N A_N$, every even rational function satisfies $C_{s,r}(hA_N)^\top Q_N = Q_N C_{s,r}(hA_N)$. Expanding the two quadratic products then gives
$$(C_{s,r} + hA_N/2)^\top Q_N (C_{s,r} + hA_N/2) = (C_{s,r} - hA_N/2)^\top Q_N (C_{s,r} - hA_N/2),$$
which proves exact Q_N-orthogonality. By construction, $C_{\mathrm{ex}} - C_{s,r} = O(z^{2s+2r})$. The Cayley resolvent identity adds one power of z, proving Equation (161); its first coefficient is nonzero under the stated hypotheses. Cancelling the greatest common polynomial factor gives the asserted intrinsic pole condition.

All coefficients of $\Pi_{s,r}$ are real. Therefore $\Pi_{s,r}(-i\xi) = \overline{\Pi_{s,r}(i\xi)}$. If either factor vanishes on the imaginary axis, so does the other, with the same multiplicity; that factor disappears in the coprime reduction. The reduced denominator consequently has no imaginary-axis zero, and the quotient has unit modulus. Its argument is twice the argument of $\Pi_{s,r}(i\xi)$, proving Equation (162). Where the $C_{s,r}$-chart is finite, the same calculation applied to $C_{s,r}(i\xi) + i\xi/2$ gives Equation (163). Finally, multiplication of Equation (158) by $h\mathcal{J}_N$ produces exactly the added even powers in $C_{s,r}(hA_N)$. □

Remark 9.19 (The Gauss–1 coincidence is exceptional)**.** *For $s = 1, r = 1$, $C_{1,1}(z) = 1 + z^2/12 = C_2(z)$, so the corrected one-stage map is exactly Gauss–2. There is no general identity "corrected Gauss–s equals Gauss–$(s+1)$." Already*

$$C_{2,1}(z) = 1 + \frac{z^2}{12} - \frac{z^4}{720}, \qquad C_3(z) = \frac{1 + z^2/10}{1 + z^2/60},$$

which agree only through the required Taylor degree and define distinct rational maps.

Proposition 9.20 (Coordinate resonances and loss of A-stability)**.** *The distinction between the gauge chart and the reduced propagator is already visible in the first nontrivial members.*

1. *For $s = 3$,*

$$C_3(z) = 6\frac{z^2 + 10}{z^2 + 60}.$$

At $z = \pm i\sqrt{60}$, the coordinate diverges, whereas for every $r \geq 1$, $\Pi_{3,r}(z) = \Pi_{3,r}(-z) = -5$ and the continued propagator is $R_{3,r}(z) = 1$.

2. *For $s = 2, r = 1$,*

$$\Pi_{2,1}(z) = 1 + \frac{z}{2} + \frac{z^2}{12} - \frac{z^4}{720}. \tag{164}$$

It has a positive real zero $\rho_\star = 10.1275149065\ldots$. Hence the reduced denominator $\Pi_{2,1}(-z)$ has a genuine pole at $z = -\rho_\star$ in the open left half-plane. The sixth-order map $R_{2,1}$ is therefore not A-stable.

Proof. For $s = 3$, $\widehat{O}_3(z) = 1/2 + z^2/120$; at its two zeros the correction in Equation (159) vanishes and direct substitution in P_3 gives -5 for both signs. For $s = 2$, $2\widehat{O}_2 = 1$, $\alpha_{2,0} = -1/720$, and Equation (164) follows. Direct polynomial root isolation gives $\Pi_{2,1}(10.1275) > 0 > \Pi_{2,1}(10.1276)$, and Newton refinement gives $\rho_\star$. Since $\Pi_{2,1}(z) - \Pi_{2,1}(-z) = z$, the numerator is nonzero at $-\rho_\star$, so the pole does not cancel. □

Remark 9.21 (Scope of the stability statement)**.** *The absence of imaginary-axis poles in Theorem 9.18 is the property relevant to the positive-energy Hamiltonian line. It does not imply left-half-plane stability for dissipative, parabolic, or split extensions. Root localization of $\Pi_{s,r}$ is therefore a separate design constraint, not a consequence of high order or exact quadratic conservation.*

Remark 9.22 (Relation with the exact-flow gauge)**.** *Equivalently,*

$$S_h^{\mathrm{exact}} - S_{s,h}^{\mathrm{G}} = h^{2s-1}\Theta_s^\star + O(h^{2s+1}).$$

Thus c_s is the leading coefficient of the difference *between the exact-flow gauge and the Gauss–s gauge. It is not, in general, the raw Bernoulli coefficient of $(z/2)\coth(z/2) - 1$. For example, at $s = 3$ the exact z^6 coefficient is $1/30240$, the Gauss–3 gauge already contributes $1/43200$, and their difference is $c_3 = 1/100800$.*

Proposition 9.23 (Stieltjes representation and Padé identification)**.** *With $u = z^2$, define*

$$F(u) = \frac{C_{\mathrm{ex}}(\sqrt{u}) - 1}{u}.$$

Then

$$F(u) = 2\sum_{k\geq 1} \frac{1}{u + 4\pi^2 k^2} = \int_{[0,1/(4\pi^2)]} \frac{\mathrm{d}\nu(\lambda)}{1 + u\lambda}, \qquad \nu = \sum_{k\geq 1} \frac{2}{4\pi^2 k^2}\delta_{1/(4\pi^2 k^2)}. \tag{165}$$

Thus F is a Stieltjes function. Moreover, $C_s(\sqrt{u})$ is the Padé approximant of type

$$\left[\left\lfloor \frac{s}{2} \right\rfloor \Big/ \left\lfloor \frac{s-1}{2} \right\rfloor\right]$$

to $C_{\mathrm{ex}}(\sqrt{u})$ at $u = 0$. Equivalently, $(C_s(\sqrt{u}) - 1)/u$ is of type $[m - 1/m - 1]$ when $s = 2m$, and of type $[m - 1/m]$ when $s = 2m + 1$.

Proof. The Mittag–Leffler expansion of coth gives the first equality in Equation (165); rewriting each summand with $\lambda_k = (4\pi^2 k^2)^{-1}$ gives the positive discrete measure. The numerator and denominator degrees of $C_s = E_s/(2\widehat{O}_s)$, expressed in u, are respectively $\lfloor s/2 \rfloor$ and $\lfloor (s-1)/2 \rfloor$. Finally, Equation (152) gives $C_{\mathrm{ex}}(\sqrt{u}) - C_s(\sqrt{u}) = O(u^s)$, which is exactly the Padé matching condition for those degrees. Division of the constant-free difference by u gives the two stated types. This places the coordinate hierarchy within classical Stieltjes–Padé theory [36]. □

Theorem 9.24 (Positive interior generation of the Gauss coordinates)**.** *Let*

$$F_s(u) = \frac{C_s(\sqrt{u}) - 1}{u}, \qquad q_s = \left\lfloor \frac{s-1}{2} \right\rfloor.$$

There exist $a_s \geq 0$, positive numbers $w_{s,k}, \lambda_{s,k} > 0$, $1 \leq k \leq q_s$, and hence

$$L_s = \mathrm{diag}(\lambda_{s,1}, \ldots, \lambda_{s,q_s}) \succ 0, \qquad b_s = (\sqrt{w_{s,1}}, \ldots, \sqrt{w_{s,q_s}})^\top,$$

such that

$$F_s(u) = a_s + \sum_{k=1}^{q_s} \frac{w_{s,k}}{1 + u\lambda_{s,k}} = a_s + b_s^\top (I + uL_s)^{-1} b_s. \tag{166}$$

The sum is absent when $q_s = 0$. Consequently C_s is generated by the positive interior system

$$(I + uL_s)\eta = b_s x, \qquad p = a_s x + b_s^\top \eta, \qquad C_s(\sqrt{u})x = x + up. \tag{167}$$

After Lanczos tridiagonalization with starting vector b_s, the same realization has a positive Jacobi matrix and repeated scalar Schur elimination gives a finite Stieltjes continued fraction.

Here positive Jacobi-chain realization *means precisely that $L_s \succ 0$, the residues $w_{s,k}$ are positive, the associated rational Stieltjes function is nonnegative on $u \geq 0$, and the Jacobi off-diagonal coefficients can be chosen positive. It does not assert that the complete augmented matrix is positive in the energy metric of the original port-Hamiltonian system.*

This positive mechanism does not generate the entire corrected hierarchy. Already

$$\frac{C_{2,1}(\sqrt{u}) - 1}{u} = \frac{1}{12} - \frac{u}{720} < 0 \qquad (u > 60), \tag{168}$$

so $C_{2,1}$ admits no realization of the positive form Equation (166).

Proof. By Proposition 9.23, F_s is a diagonal or subdiagonal Padé approximant of the Stieltjes function F. The classical Stieltjes–Padé theorem gives real negative simple poles and positive residues; a diagonal member may additionally have a nonnegative constant term at infinity [36]. Rewriting every pole as $-1/\lambda_{s,k}$ gives Equation (166). This is the finite positive-measure realization of a rational Stieltjes function.

Eliminating η in Equation (167) gives $p = F_s(u)x$, hence the last identity. Orthogonal Lanczos reduction of L_s with starting vector b_s preserves the scalar resolvent and produces a symmetric tridiagonal Jacobi matrix; successive block elimination of its last coordinate yields the Stieltjes continued fraction [35]. The corresponding positive-network interpretation belongs to classical network synthesis [1]. Finally, the right-hand side of Equation (166) is nonnegative for every $u \geq 0$, whereas direct substitution of $C_{2,1}$ gives Equation (168). The claimed obstruction follows. □

Remark 9.25 (Two distinct notions of interior generation)**.** *Theorem 9.10 uses a nondegenerate alternating interior block and can generate every finite-rank cap two-form, but it may be indefinite and is designed from the target gauge. By contrast, Theorem 9.24 uses a positive constitutive interior solve and gives the canonical Jacobi chain behind Gauss–Padé. The failure in Equation (168) shows that the Taylor-corrected central hierarchy cannot simply be reinterpreted as the same passive chain with extra stages. A field-generated realization of those corrections must either use a different rational hierarchy or admit indefinite/nonpassive internal structure.*

Remark 9.26 (A coefficientwise sign conjecture)**.** *Exact arithmetic for* $1 \le s \le 12$ *and* $0 \le j \le 12$ *gives*

$$(-1)^{s+j+1}\alpha_{s,j} > 0. \tag{169}$$

If proved for all s, j*, this would remove the non-annulation hypothesis from Corollary 9.17 and Theorem 9.18. The Stieltjes representation makes Equation (169) natural, but standard real-axis Padé error inequalities do not by themselves establish this coefficientwise statement. We therefore retain it as a conjecture, and every exact-order claim continues to state the hypothesis* $\alpha_{s,r} \neq 0$*.*

Corollary 9.27 (The Gauss–2 member)**.** *For* $s = 2$*,*

$$S^{\Theta}_{2,h} = \frac{h}{12} Q_N \mathcal{J}_N Q_N + h^3 \Theta. \tag{170}$$

Writing

$$D_2 = I - \frac{h}{2}A_N + \frac{h^2}{12}A_N^2, \qquad N_2 = I + \frac{h}{2}A_N + \frac{h^2}{12}A_N^2, \qquad E_h = h^4 \mathcal{J}_N \Theta,$$

the propagator is

$$M_{2,\Theta,h} = (D_2 + E_h)^{-1}(N_2 + E_h). \tag{171}$$

It is the fourth-order example used below.

Proof. For $P_2(z) = I + z/2 + z^2/12$, $E_2(z) = I + z^2/12$ and $\widehat{O}_2(z) = I/2$. Substitution in Equations (138) and (142) gives the stated formulas. □

Remark 9.28 (What is and is not new in the line example)**.** *The case* $\Theta = 0$ *is only Gauss–s applied to the 2012 spatial reduction and lies within the architecture of [18]. The difference is the lifted* Q_{tx} *term. It preserves the proved order and power but, under the commutator hypothesis, destroys both the scalar-coefficient Runge–Kutta method-of-lines representation and symplecticity. Thus the construction does not relabel Gauss as space–time; it contains the whole Gauss hierarchy as the separable zero-mixed-gauge family, realizes every deformation through the raw cylinder system, and gives a falsifiable algebraic criterion for leaving the broader scalar functional-calculus subclass. The one-term classification also exposes a sharp tradeoff: its unique higher-order coefficient is central. The two-term family in Theorem 9.16 removes that apparent obstruction by retaining order* $2s+2$ *with an arbitrary noncommuting next coefficient. High-order energy-preserving methods exist in other forms [25, 10, 16]; the specific contribution here is the mixed cylinder realization and the nonseparability criterion. Numerical advantage still requires a selector; below, modal dispersion provides one such test.*

Energy currents and multisymplectic forms. Polarization of the quadratic energy law gives a symmetric bilinear current. A multisymplectic form formula is instead an alternating two-form evaluated on two variational solutions [2, 27]. Gauss collocation may preserve both because the Cooper condition preserves quadratic invariants, but the two tensor parities must not be identified. In the present theory, symmetric power balance holds throughout the compatible affine family, while the alternating temporal symplectic law holds exactly on $\mathcal{L}$.

10 Consequences and discriminating computational validation

10.1 Classification tables

Table 1: Properties of a fixed, h-independent one-step linear gauge for nonsingular J, Q. Step-dependent odd gauges such as Equation (139) are not covered by the order column.

Gauge	Exact energy	Symplectic	Order
$S = 0$	yes	yes	two
$S \in \mathcal{L} \setminus \{0\}$	yes	yes	one
$S \notin \mathcal{L}$	yes	no	one

Table 2: Nonlinear two-point methods: generic structural properties.

Method	Exact prescribed energy	Symplectic generically	Affine-covariant construction
AVF discrete gradient	yes	no	yes
Gonzalez discrete gradient	yes	no	no, metric-dependent
Implicit midpoint	only for quadratic H	yes	yes

10.2 Verification on a stacked space–time polynomial cell

We first test the stacked, nonsurjective differential rather than the genuine total-complex map or a time-stepping reduction. Take

$$E_t = \mathcal{P}_3, \qquad E_x = \mathcal{P}_4,$$

with monomial coefficients on $[0,1]^2$, exact L^2 product pairings, and

$$D_{\mathrm{st}} z = ((\partial_t \otimes I)z, (I \otimes \partial_x)z).$$

Then $\dim E_{\mathrm{st}} = 20$, $\dim F_{\mathrm{st}} = 31$, and the computed rank is 19, exactly as predicted by Equation (31); hence $\dim \mathcal{G}_{\mathrm{st}} = 19 \cdot 18/2 = 171$.

Starting from the exact product representative, we add unit-Frobenius gauges supported respectively in the temporal, spatial, mixed, and temporal–mixed cross sectors of Equation (37). The matrix residual

$$\varepsilon_{\mathrm{st}} = \left\| M_{\mathrm{st}} D_{\mathrm{st}} + (M_{\mathrm{st}} D_{\mathrm{st}})^{\top} - B_{\mathrm{st}} \right\|_F \tag{172}$$

and the alternating self-contraction are reported in Table 3.

This check confirms two claims that cannot be seen in a method-of-lines experiment: the active rank is smaller than $\dim F_{\mathrm{st}}$, and a genuinely mixed nonseparable gauge can satisfy the same global Stokes identity as the product representative.

10.3 The 2012 ideal line: reconstruction, active stages, and design

We use the physical parameters and complementary boundary causality of the ideal-line experiment in [20]:

$$\ell = 2, \qquad C = 3, \qquad L = 2,$$

Table 3: Compatibility residuals on the $\mathcal{P}_3 \otimes \mathcal{P}_4$ space–time cell.

Gauge sector	$\varepsilon_{\mathrm{st}}$	$\lvert\omega(z,z)\rvert$
zero representative	5.93×10^{-16}	0
temporal	6.71×10^{-16}	5.55×10^{-17}
spatial	6.71×10^{-16}	0
mixed	6.71×10^{-16}	3.99×10^{-18}
temporal–mixed cross	5.93×10^{-16}	0

with $N = 8$. In the shifted $L^2(0,\ell)$-orthonormal Legendre basis, the exact rectangular power chart is particularly transparent: $M_N = [I_N; 0]$, while D_N and both endpoint traces are evaluated exactly. The residual in Equation (118) is 3.97×10^{-13}. Eliminating the degree-N radical under the causality Equation (122) gives Equation (135), with

$$\left\|\mathcal{J}_N + \mathcal{J}_N^{\top}\right\|_F = 3.08 \times 10^{-13}, \quad \left\|C_N - B_N^{\top}\right\|_F = 7.66 \times 10^{-14}, \quad \left\|N_N + N_N^{\top}\right\|_F = 0.$$

The stronger blockwise test of Theorem 9.2 and Equation (136) is shown in Table 4. Every square open-system block is reconstructed from (M_N, D_N) and the four endpoint traces.

Table 4: Rectangular-to-square reconstruction of the open line.

Quantity	residual or condition number
polarized rectangular Stokes block	5.61×10^{-13}
recovery of $\mathcal{J}_N$	1.10×10^{-14}
recovery of B_N	7.35×10^{-16}
recovery of $B_N^{\top}$ from the output trace	3.26×10^{-15}
recovery of N_N	1.11×10^{-16}
minimum singular value of each endpoint reconstruction	4.54×10^{-1}
$\kappa_2(\mathcal{R}_N)$	1.42×10^{1}

The transmission-line computation is organized as a falsifiable validation of the rectangular Stokes theorem, the volume-to-cap pullback, exact trace-generated realization, independence of the raw gauge fixing, the Cayley bijection and its conditioning, active-stage Gauss condensation, the open power law, the exact order dichotomy and its unique order-raising coefficient, the scalar-functional-calculus obstruction, dispersion-based gauge selection, and cancellation after solving two coupled cells. We choose a rank-six alternating Θ coupling distinct q- and p-Legendre modes. Its scale is fixed in energy-orthonormal coordinates:

$$\left\|Q_N^{-1/2}\Theta Q_N^{-1/2}\right\|_F = \left\|Q_N^{-1/2}(Q_N\mathcal{J}_N Q_N/12)Q_N^{-1/2}\right\|_F. \tag{173}$$

Equivalently, this is the norm on covariant two-forms induced by Q_N, and is invariant under a change of state coordinates. For the reduction test we use the following numerical section:

$$\sigma_x = \mathcal{D}_x^{\top}(\mathcal{D}_x\mathcal{D}_x^{\top})^{-1}.$$

It is inserted in Equation (127) and is not used in the time-step solve and therefore tests Proposition 9.9 independently; in coordinates the lifted form is $\Omega_{\mathrm{st}} = \mathcal{T}_N^{\top}\Theta\mathcal{T}_N$. The cap dimension is 16, so surjectivity predicts all $\binom{16}{2} = 120$ cap gauges to be reachable. The residuals are reported in Table 5.

Table 5: Volume-to-cap pullback on the degree-eight transmission line.

Quantity	residual or rank
$\|\mathcal{T}_N\mathcal{L}_N - I\|_F$	1.29×10^{-14}
$\|\Omega_{\mathrm{st}} + \Omega_{\mathrm{st}}^\top\|_F$	4.77×10^{-14}
$\|\mathcal{L}_N^\top\Omega_{\mathrm{st}}\mathcal{L}_N - \Theta\|_F/\|\Theta\|_F$	5.73×10^{-15}
$\operatorname{rank}\mathcal{T}_N$	16
reachable cap-gauge dimension	120

The preceding pullback is followed by an exact raw-realization test. At $h = 0.1$, set $S_h = S_{2,h}^{\Theta}$ and assemble the full raw system Equation (59) with $\Omega_h = \mathcal{T}_N^\top S_h \mathcal{T}_N$. The raw potential has dimension 36; the effort remains in the full rectangular space $(\mathcal{P}_8)^2$ of dimension 18; and the two endpoint conditions $e_q(0) = e_p(\ell) = 0$ are imposed explicitly. Since $\ker\mathcal{T}_N$ has dimension 20, including the final cap and the null-space multiplier gives a 90×90 system. Its blocks use D_N, M_N, the endpoint traces, and Ω_h; neither $\mathcal{J}_N$ nor a cap propagator is inserted. After solving the system for the sixteen coordinate initial states, its x_1-block is compared with Equation (171).

In coordinates, the input trace in Equation (134) is

$$\mathcal{C}_N = \begin{bmatrix} r_0^\top & 0 \\ 0 & r_\ell^\top \end{bmatrix}.$$

If the columns of N_T span $\ker\mathcal{T}_N$, the implemented rectangular cell is

$$\begin{bmatrix} \mathcal{T}_N & 0 & -I & 0 \\ 0 & -h\mathcal{K}_N & I & 0 \\ \Omega_h & \mathcal{T}_N^\top\mathcal{M}_N^* & -\frac{1}{2}\mathcal{T}_N^\top Q_N & N_T \\ N_T^\top & 0 & 0 & 0 \\ 0 & \mathcal{C}_N & 0 & 0 \end{bmatrix} \begin{bmatrix} z \\ e \\ x_1 \\ \lambda \end{bmatrix} = \begin{bmatrix} -I \\ I \\ \frac{1}{2}\mathcal{T}_N^\top Q_N \\ 0 \\ 0 \end{bmatrix} x_0. \tag{174}$$

The square map $\mathcal{J}_N$ is itself recovered only during elimination:

$$\mathcal{J}_N = \mathcal{K}_N \begin{bmatrix} \mathcal{M}_N^* \\ \mathcal{C}_N \end{bmatrix}^{-1} \begin{bmatrix} I \\ 0 \end{bmatrix}. \tag{175}$$

Table 6: Exact realization by the raw trace-generated line cell.

Quantity	value
raw block-system residual	6.80×10^{-15}
relative error of the condensed cap map	1.92×10^{-16}
relative change under a different right inverse	1.05×10^{-14}
cap-map change under a different raw gauge fixing	1.17×10^{-16}
alternative raw-system residual	6.73×10^{-15}
null-space multiplier norm	7.75×10^{-16}
recovery of $\mathcal{J}_N$ from rectangular blocks	1.10×10^{-14}
condition number of the raw system	1.53×10^{2}

This test has the precise scope of Theorem 4.14: the prescribed cap gauge is encoded in the raw volume matrix and recovered by elimination. It is not presented as selection of that gauge by independent interior dynamics. The additional gauge-fixing test solves the complete

raw system with a nonorthogonal complement of $\ker \mathcal{T}_N$; the physical cap map is unchanged to roundoff.

The genuinely interior test is instead the polynomial Gauss cell of Theorem 8.8. We assemble its sn-dimensional stage block for $s = 1, \dots, 4$, eliminate the stages, and only then recover the cap gauge by Equation (71). The results are in Table 7; the largest energy defect is 2.69×10^{-15}.

Table 7: Active temporal-stage elimination at $h = 0.1$, $n = 16$.

s	sn	stage residual	map–Padé	induced gauge	κ_2
1	16	4.73×10^{-16}	1.77×10^{-16}	1.88×10^{-14}	1.608
2	32	9.87×10^{-16}	2.59×10^{-16}	2.90×10^{-14}	2.976
3	48	1.49×10^{-15}	2.45×10^{-16}	2.31×10^{-14}	3.660
4	64	2.00×10^{-15}	1.35×10^{-15}	4.75×10^{-14}	3.805

The Cayley coordinate is first checked independently at the moderate step $h = 0.1$. A prescribed alternating gauge is mapped to $O(Q_N)$, inverted by Equation (71), and mapped forward again. The gauge and map round trips have relative errors 6.07×10^{-13} and 1.93×10^{-16}; the energy-orthogonality defect is 2.34×10^{-15}. Applying the same algebraic inverse to $\exp(hA_N)$ and reconstructing the flow gives 2.99×10^{-16} relative error.

That test does not conceal the singular conditioning of the inverse chart near I. In Table 8, the reference exact-flow gauge is evaluated directly by the Bernoulli series in Equation (74). The "naive" column instead forms $\exp(hA_N) - I$ and applies Equation (71). Its rapidly growing error is predicted by Remark 5.5, while the map reconstructed from the direct gauge remains at roundoff.

Table 8: Stable exact-flow gauge evaluation near the excluded identity.

h	relative naive gauge error	error $h^3/\epsilon_{\mathrm{mach}}$	stable map error
10^{-2}	2.36×10^{-10}	1.062	1.43×10^{-16}
10^{-4}	2.60×10^{-4}	1.169	1.47×10^{-16}
10^{-6}	2.62×10^{2}	1.181	1.55×10^{-16}
10^{-8}	1.31×10^{8}	0.591	1.11×10^{-16}

The same form is lifted to the cylinder by Equation (125). At $h = 0.04$, the open balance is tested for midpoint, the second-order mixed deformation $S_h = h\Theta$, and the fourth-order deformation Equation (170).

Table 9: Direct rectangular transmission-line cap–mantle balance.

Gauge	ΔH_N	$h\, y^\top u$	residual
midpoint	$2.67994318 \times 10^{-2}$	$2.67994318 \times 10^{-2}$	5.13×10^{-16}
mixed, order two	$2.67401314 \times 10^{-2}$	$2.67401314 \times 10^{-2}$	6.31×10^{-16}
Gauss–2 plus mixed, order four	$2.74765368 \times 10^{-2}$	$2.74765368 \times 10^{-2}$	2.71×10^{-16}

For the homogeneous line, the zero-mixed member of Equation (170) is Gauss–2. At $h = 0.05$, its relative commutator with A_N is 6.92×10^{-17}. The mixed deformation has energy defect 1.51×10^{-15}, relative commutator 2.59×10^{-6}, and relative symplectic defect 1.20×10^{-6}. Refinement gives Table 10; the one-step error has the expected fifth-power scaling, while the commutator remains decisively above roundoff.

Table 10: Fourth-order mixed deformation of Gauss–2 on the closed line.

h	relative one-step error	one-step error slope	relative $\|[M_{\Theta,h}, A_N]\|_F$
0.1000	8.9501×10^{-3}	—	7.3650×10^{-5}
0.0500	3.3624×10^{-4}	4.734	2.5947×10^{-6}
0.0250	1.1167×10^{-5}	4.912	8.3450×10^{-8}
0.0125	3.5483×10^{-7}	4.976	2.6264×10^{-9}

To distinguish a one-step slope from convergence of the integrator, we also fix the final time $T = 0.5$, compose the map, and compare it with $\exp(TA_N)$. For each s, the mixed coefficient Θ is held fixed and the perturbation is exactly $h^{2s-1}\Theta$. The final refinement slopes in Table 11 approach the proved global orders $2, 4, 6$; the largest one-step energy defect over the table is 3.08×10^{-15}.

Table 11: Global convergence of the stage-induced, mixed Gauss–s hierarchy at $T = 0.5$. The four error columns correspond to $4, 8, 16, 32$ steps.

s	expected order	$n = 4$	$n = 8$	$n = 16$	$n = 32$	final slope
1	2	6.398×10^{-1}	2.870×10^{-1}	8.393×10^{-2}	2.179×10^{-2}	1.946
2	4	9.816×10^{-2}	8.058×10^{-3}	5.368×10^{-4}	3.407×10^{-5}	3.978
3	6	4.462×10^{-3}	8.240×10^{-5}	1.340×10^{-6}	2.115×10^{-8}	5.986

The preceding coefficient is noncommuting and therefore cannot equal the exceptional value in Equation (146). The same experiment is now repeated with $\Theta = \Theta_s^\star$. The observed orders in Table 12 are $4, 6, 8$, and the largest energy defect is 3.27×10^{-15}. For $s = 1$, the relative difference between the raised map and Gauss–2 at $h = 0.1$ is 2.42×10^{-16}, confirming the exact identity in Theorem 9.14.

Table 12: Global convergence of the unique order-raising coefficient in the fixed one-term ansatz at $T = 0.5$. The error columns correspond to n time steps.

s	predicted order	$n = 4$	$n = 8$	$n = 16$	$n = 32$	$n = 64$	final slope
1	4	9.950×10^{-2}	8.184×10^{-3}	5.454×10^{-4}	3.462×10^{-5}	2.172×10^{-6}	3.994
2	6	1.335×10^{-2}	2.689×10^{-4}	4.466×10^{-6}	7.084×10^{-8}	1.111×10^{-9}	5.995
3	8	1.027×10^{-3}	4.985×10^{-6}	2.053×10^{-8}	8.123×10^{-11}	3.189×10^{-13}	7.993

The order statement is not inferred from these slopes. To test the new coefficient in Equation (147) directly, we use the unit harmonic oscillator $A = J_2$ and 90-digit arithmetic. After dividing the one-step matrix error by h^{2s+3}, its component along J_2 must converge to $(-1)^{s+1}d_s$. At $h = 0.025$ the recovered values in Table 13 already agree to about 3.5×10^{-4} relatively. The full normalized remainder decreases linearly with h, exactly as the $O(h^{2s+4})$ term predicts.

Table 13: Direct local asymptotic test of the exact coefficient d_s.

s	d_s	predicted J_2 coefficient	recovered coefficient at $h = 0.025$
1	$-1/720$	$-1.388888889 \times 10^{-3}$	$-1.388403230 \times 10^{-3}$
2	$1/30240$	$-3.306878307 \times 10^{-5}$	$-3.305724437 \times 10^{-5}$
3	$-1/2268000$	$-4.409171076 \times 10^{-7}$	$-4.407653975 \times 10^{-7}$

We next retain $\Theta_s^\star$ and add the same fixed noncommuting line two-form $h^{2s+1}\Xi$ for

$s = 1, 2, 3$. The global orders remain $4, 6, 8$, but the commutators in Table 14 are nonzero at resolved step sizes. This is the experimental distinction absent from the one-term exceptional branch: high order and departure from scalar functional calculus now coexist. The largest one-step energy defect in the experiment is 3.11×10^{-15}.

Table 14: Two-term high-order noncommuting hierarchy on the transmission line. Errors are global at $T = 0.5$; commutators are reported at $n = 16$, before the higher-stage values reach roundoff.

s	exact order	error, $n = 32$	final slope	relative commutator, $n = 16$
1	4	3.407×10^{-5}	3.994	2.533×10^{-7}
2	6	7.071×10^{-8}	5.995	2.474×10^{-10}
3	8	8.120×10^{-11}	7.993	2.416×10^{-13}

For $s = 3$, binary64 eventually masks the noncommuting signal. We therefore repeat the one-step calculation in 90-digit decimal arithmetic on the following fully specified four-dimensional Hamiltonian test:

$$\mathcal{J} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{bmatrix}, \qquad Q = \operatorname{diag}(1, 2, 3, 5), \qquad A = \mathcal{J}Q, \qquad \Xi = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}. \tag{176}$$

The common block is evaluated directly as

$$C_3(hA) + \frac{(hA)^6}{100800} + h^8 \mathcal{J}\Xi, \qquad C_3(z) = \frac{1}{2}\left(I + \frac{z^2}{10}\right)\left(\frac{I}{2} + \frac{z^2}{120}\right)^{-1},$$

using dense decimal Gaussian elimination with partial pivoting. Thus the experiment requires no hidden random matrix or fitted coefficient. The theorem predicts

$$[M_h, A] = -h^9[\mathcal{J}\Xi, A]A + O(h^{10}).$$

Table 15 verifies both the exponent and the full leading matrix after division by h^9; this is independent of the global-order slope above.

Table 15: Multiprecision resolution of the $s = 3$ noncommutation law.

h	$\|[M_h, A]\|_F$	observed exponent	relative error in the leading matrix
0.050000	4.044×10^{-11}	–	8.436×10^{-2}
0.025000	7.901×10^{-14}	8.99967	4.223×10^{-2}
0.012500	1.543×10^{-16}	8.99992	2.112×10^{-2}
0.006250	3.014×10^{-19}	8.99998	1.056×10^{-2}

The reduced-polynomial checks are independent of the small-step order tests. At the C_3 chart resonance the computed $|\widehat{O}_3|$ is 1.11×10^{-16}, while $|\Pi_{3,1} + 5| = 4.44 \times 10^{-16}$ and the continued map differs from one by 1.78×10^{-16}. Root computation for Equation (164) gives the genuine left-half-plane pole -10.12751490652, with denominator residual 1.78×10^{-14} and numerator magnitude 10.1275. On 4001 points $\xi \in [-100, 100]$, the largest defect in $|R_{2,1}(i\xi)| = 1$ is 4.44×10^{-16}. These three tests separate a removable coordinate resonance, a true propagator pole, and the protected oscillatory axis.

Finally, exact `Fraction` arithmetic evaluates all 156 coefficients $\alpha_{s,j}$ with $1 \le s \le 12$, $0 \le j \le 12$. No zero or sign failure relative to Equation (169) occurs. This is reported only as finite evidence for the conjecture, not as its proof.

The interior-generation assertions directly target the two new results. For a rank-four random alternating cap change in dimension six, the four-dimensional construction in Theorem 9.10 has zero skew defect, Schur error 8.02×10^{-16}, and eliminated-response error 1.83×10^{-15}. For Gauss–3, the one-mode data

$$a_3 = 0, \qquad w_{3,1} = \frac{1}{12}, \qquad \lambda_{3,1} = \frac{1}{60}$$

reproduce both $F_3(u) = 5/(u+60)$ and C_3 to roundoff on $u \in [0, 120]$. At $u = 61$, the corrected value in Equation (168) is exactly $-1/720$, so the positive-realization obstruction is not a floating-point inference.

Choosing the second central corrector $\Xi_s^\star = d_s \Omega_N A_N^{2s+2}$ gives observed global orders six and eight for $s = 1, 2$, respectively; see Table 16. These are the first two nontrivial levels of the recursive exact-gauge expansion in Corollary 9.17.

Table 16: Second central correction on the transmission line.

s	proved lower bound	error, $n = 32$	final slope
1	6	7.084×10^{-8}	5.995
2	8	1.522×10^{-10}	7.994

The affine freedom can also be used as a design space. We minimize the relative root-mean-square error of the eight positive modal frequencies within the one-dimensional family $S_{2,h}^{\alpha\Theta} = S_{2,h}^{\mathrm{G}} + \alpha h^3 \Theta$, using only the small reference step $h_\star = 1/80$. This gives $\alpha_\star = 57.3601$, which is then frozen. The tests in Table 17 use only unseen, coarser steps.

Table 17: Dispersion selection in a fixed one-parameter mixed family.

h	Gauss RMS phase error	selected error	reduction	κ_2 Gauss	κ_2 selected
0.05	3.605×10^{-4}	1.143×10^{-4}	68.28%	1.202	1.202
0.10	4.864×10^{-3}	1.875×10^{-3}	61.45%	1.448	1.443
0.15	1.845×10^{-2}	9.589×10^{-3}	48.03%	1.861	1.941

The selected map remains exactly energy preserving; only its phase is optimized. The improvement persists away from the fitting step and the linear-system condition number changes mildly. Repeating the norm calculation after a fixed nonorthogonal coordinate change gives a relative difference below displayed precision, confirming the invariance of Equation (173). This experiment does not claim universal superiority of the selected gauge, but it does answer a sharper question: the mixed degree of freedom can correct dispersion without sacrificing the exact power identity or the proved temporal order.

Finally, two degree-eight cells are solved simultaneously with the power-preserving interface constraints of Proposition 9.8, the fourth-order mixed gauge, and $R_N = 0.02I$. The algebraic solve residual is 5.06×10^{-16}, the computed interface power is 3.33×10^{-16}, and

$$\Delta H_{N,1} + \Delta H_{N,2} = 2.75921723 \times 10^{-2},$$

while external supply minus dissipation is

$$3.29176951 \times 10^{-2} - 5.32552281 \times 10^{-3} = 2.75921723 \times 10^{-2},$$

with residual 1.78×10^{-15}. These calculations test exactly the structural chain asserted in this paper: the rectangular Stokes identity produces the open Dirac chart; active temporal stages generate the Gauss cap gauge; the mixed volume form realizes its prescribed perturbation; contracted power is unchanged; the hierarchy attains orders exactly $2s$ away from the unique one-term coefficient $\Theta_s^\star$, which raises the order to exactly $2s+2$; a second noncommuting term retains that higher order and leaves the scalar functional-calculus subclass; the central series raises the order further; a fixed gauge can reduce modal phase error; and internal supplies cancel in a solved, dissipative two-cell system. Additional assertions verify a rank-minimal skew Schur generator, the positive one-mode realization of C_3, and the sign obstruction $(C_{2,1}(\sqrt{u})-1)/u < 0$ beyond $u = 60$.

All values are generated by the accompanying script `verify_transmission_line_v12.py`. Its 28 assertion groups use explicit tolerances and convergence thresholds; the program terminates with an `AssertionError` if any reported structural claim fails.

10.4 Numerical verification of the linear locus

We verify the identities in double precision on the two-mode system of Example 6.6, with

$$Q = \mathrm{diag}(I_2, 2I_2), \qquad S_{\mathrm{in}} = \mathrm{diag}(0.3J_2, -0.2J_2), \qquad S_{\mathrm{out}} = \begin{bmatrix} 0 & I_2 \\ -I_2 & 0 \end{bmatrix}.$$

Here $S_{\mathrm{in}} \in \mathcal{L}$ and $S_{\mathrm{out}} \notin \mathcal{L}$. At $h = 0.07$, the Frobenius residuals

$$\varepsilon_H = \left\| M_S^\top Q M_S - Q \right\|_F, \qquad \varepsilon_\Omega = \left\| M_S^\top \Omega M_S - \Omega \right\|_F$$

are shown in Table 18. Energy is at roundoff for every gauge, whereas the symplectic residual separates the locus.

Table 18: One-step structural residuals at $h = 0.07$.

Gauge	ε_H	ε_Ω
$S = 0$	5.14×10^{-16}	2.94×10^{-16}
S_{in}	2.90×10^{-16}	1.58×10^{-16}
S_{out}	7.19×10^{-16}	9.82×10^{-3}

For the off-locus gauge, decreasing h from 0.1 to 0.0125 gives

$$\frac{\varepsilon_\Omega}{h^2} = 2.0074,\ 2.0019,\ 2.0005,\ 2.0001,$$

confirming Theorem 6.3. To test order, we integrate to $T = 1$ from $x_0 = (0.7, -0.2, 0.4, 0.9)^\top$ and compare with $\exp(A)x_0$. The last two mesh refinements in Table 19 show the predicted orders.

Table 19: Global error and observed order at $T = 1$.

Gauge	error, $h = 1/80$	error, $h = 1/160$	observed order
$S = 0$	1.0302×10^{-4}	2.5757×10^{-5}	1.9999
S_{in}	5.7094×10^{-3}	2.8350×10^{-3}	1.0100
S_{out}	2.5020×10^{-2}	1.2547×10^{-2}	0.9957

These checks are algebraic rather than qualitative long-time plots: any contrary result falsifies either the implementation or one of the stated nondegeneracy assumptions.

10.5 Summary of established results and hypotheses

The conjectural statements motivating this work are resolved as follows.

1. For an arbitrary linear map $D : E \to F$, the unrestricted active gauge is exactly $\Lambda^2(E/\ker D)^*$: Theorem 2.2.
2. For the genuine total complex, symmetric doubling gives $\Lambda^2(U_q \oplus V_q)^*$; imposing complementary degree reduces this to $U_q^* \otimes V_q^*$: Theorem 4.3. The tensor pairing and both oriented boundary terms, including their Koszul signs, are explicit in Equations (16) and (17). This includes cancellations between bidegrees through the actual total-complex kernels.
3. For the distinct stacked tensor-cylinder map, $\ker D_{\mathrm{st}} = K_t \otimes K_x$, the rank is $n_t n_x - k_t k_x$, and the gauge decomposes into temporal, spatial, and mixed sectors through a canonical filtration: Theorem 4.6, Proposition 4.7, and Corollary 4.8.
4. In every fixed regular reduction chart, the volume-to-cap map is L_h^*, and its image, kernel, and dimensions are exactly Equation (40)–Equation (44). The canonical mixed part reduces through $(\pi_{tx} L_h)^*$, with no chosen splitting: Corollary 4.12.
5. For a surjective trace, the gauge-bearing weak cell Equation (59) realizes exactly the prescribed cap gauge in Equation (57), provided the stated map block is invertible. This is an encoding-and-recovery result, not a claim of interior selection. The fixed-reconstruction theorem is also the tangent map of a gauge-dependent reduction: Proposition 4.16.
6. The complete temporal linear gauge is $\Lambda^2 V^*$; which members arise from a selected volume model is determined by $\operatorname{im} \mathcal{R}_{L_h}$: Theorems 4.10 and 5.1.
7. For nonsingular J, Q, the temporal family gives a bijective, constitutively shifted Cayley coordinate on $\{M \in O(Q) : M - I \text{ invertible}\}$. Its inverse and the exact-flow gauge are Equations (71) and (73); the inverse coordinate is condition-sensitive near I, as stated in Remark 5.5, and the symplectic subfamily maps to $O(Q) \cap Sp(\Omega)$.
8. The gauge-invariant power balance is exact, including ports and dissipation: Corollaries 5.2 and 9.4.
9. The midpoint rule is uniquely selected by endpoint symmetry, midpoint localization, and—for nonsingular J, Q—order two: Proposition 5.6.
10. Symplecticity does not select the zero gauge. Its exact locus is Equation (76); generically it has dimension m, and in two dimensions it is the entire gauge: Theorem 6.4 and Corollary 6.5.
11. AVF is uniquely selected only in the stated universal segment-measure class: Theorem 7.2.
12. For invertible stage matrices, the stage-pairing existence criterion, dimension, and relation to $R(\infty)$ are given by Theorem 8.2; the singular case remains the range problem stated in Remark 8.3.
13. Among collocation methods, local compatible pairing is equivalent to Gauss–Legendre collocation: Theorem 8.5.
14. Conversely, the active temporal cell $\mathcal{P}_s \to \mathcal{P}_{s-1}$ generates the Gauss–s cap map by eliminating sn interior stage variables; the even–odd Padé factorization gives its two-point gauge directly: Proposition 8.7 and Theorem 8.8.

15. The global tensor Stokes equations force a local diagonal pairing to factor, and maximal polynomial exactness then gives the product Gauss rule; this is not a uniqueness statement among nonlocal gauges: Theorem 9.5.

16. A genuine finite-dimensional Dirac claim requires isotropy and maximality. These are proved in Proposition 9.1, including an antisymmetric feedthrough; the temporal effort is correctly identified as a constitutive closure.

17. An invertible rectangular effort-and-input reconstruction turns the polarized Stokes identity into a square open Dirac representation, forcing skew interconnection, input/output duality, and skew feedthrough: Theorem 9.2. For the polynomial line, the required reconstruction is proved invertible in Lemma 9.12.

18. A mixed gauge can preserve the full cylinder identity while having zero pure temporal and spatial pullbacks; the polynomial cell check in Table 3 exhibits this possibility directly.

19. On the rectangular 2012 transmission-line complex, the cap trace is onto and every cap two-form has the explicit pure-mixed lift Equation (125). The hierarchy Equation (139) retains exact energy and reversibility. It has exactly order $2s$ for every $\Theta \neq \Theta_s^\star$, whereas the unique fixed coefficient in the one-term ansatz Equation (146) raises the order to exactly $2s+2$. The two-term family Equation (154) retains that order with an arbitrary next coefficient, and the central choice Equation (156) raises it by two more. Moreover, Equation (143) detects departure from the scalar functional calculus of the semidiscrete line.

20. The central truncation has the intrinsic reduced form $R_{s,r} = \Pi_{s,r}(-z)^{-1}\Pi_{s,r}(z)$: Theorem 9.18. Coordinate poles of C_s are removable $+1$ resonances, the reduced map has no pole on the imaginary axis, but it need not be A-stable; the explicit $(s,r) = (2,1)$ member has a left-half-plane pole: Proposition 9.20. The exact coordinate is a Stieltjes function in z^2, and C_s is its stated Padé approximant: Proposition 9.23. The strict sign law for every $\alpha_{s,j}$ remains the explicitly labelled conjecture Remark 9.26.

21. At the unrestricted algebraic level, every rank-$2q$ change of cap gauge is generated by eliminating exactly $2q$ auxiliary alternating modes, and no smaller generator of that block form is possible: Theorem 9.10. This sharp existence-and-dimension statement is distinct from a field-selection mechanism. The stronger positive Jacobi-chain mechanism generates every Gauss coordinate, but not the corrected $(2,1)$ member: Theorem 9.24. Thus algebraic generation, passive generation, and selection by an independent field cell are three distinct statements. Tables 4, 6 to 8 and 11 to 17 verify the three reductions, stable coordinate evaluation, the direct asymptotic coefficient, the one- and two-term order branches, the multiprecision noncommutation exponent, and a concrete dispersion gain; the solved two-cell test verifies interface power.

11 Discussion and limitations

Four layers should remain distinct in future extensions.

1. *Stokes compatibility* is an affine constraint on a pairing. Its homogeneous direction is alternating on a quotient space.

2. *Dirac maximality* concerns a subspace of flows and efforts equipped with a symmetric split pairing. Isotropy alone is insufficient; the dimension or orthogonal-complement condition must also be checked.

3. *Constitutive closure* inserts the Hamiltonian through a continuous or discrete gradient. It supplies energy balance but is not itself a proof of maximal isotropy.
4. *Boundary reduction* composes the preceding layers and eliminates interior variables. Only after a regular reconstruction L_h has been established is the reduced gauge $L_h^*\omega$; singular interior blocks require a reduction of relations.

Compatible gauges should also be distinguished by their origin.

1. A *geometric or premetric gauge* depends only on the differential, traces, orientation, and approximation spaces. The local Gauss stage pairing is of this type.
2. A *constitutively adapted gauge* is obtained only after J, Q, and h are supplied. The induced Gauss cap gauge Equation (99) and the exact-flow gauge Equation (73) belong to this class.
3. A *design gauge* uses the remaining compatible freedom to optimize dispersion, conditioning, sparsity, or another numerical objective. The one-parameter mixed experiment is an example, not a universal selector.

All three classes satisfy the same affine Stokes constraint, but only the first can be attributed to the discrete geometry alone.

The central rational hierarchy is tailored to Hamiltonian generators, not to the whole left half-plane. Its reduced polynomial form removes the artificial poles of the C_s-coordinate and proves unit amplification on the oscillatory axis, but Proposition 9.20 shows that high order and exact quadratic conservation do not imply A-stability. Consequently dissipative extensions require an independent root analysis of $\Pi_{s,r}$. Likewise, the Stieltjes–Padé identification is proved, whereas the coefficientwise sign and non-annulation law Equation (169) is not; all hierarchy orders beyond the computed levels retain the stated conditional wording.

The cylinder and double-complex results are finite-dimensional discrete statements. They do not by themselves prove convergence to an infinite-dimensional field theory. Such a result requires approximation, stability, trace, and compactness estimates for the selected trial spaces. Recent space–time FEEC work combines fourth-order Gauss–Legendre time integration with a spatial finite-element complex and proves a local multisymplectic conservation law for nonlinear wave equations [7]. That PDE convergence and local variational statement is complementary to, and stronger in a different direction than, the exact finite-dimensional quadratic cap-power law proved here. Descriptor systems and state-dependent mass matrices require additional constraint hypotheses; discrete-gradient pairs provide one relevant extension [30]. Likewise, the compatible graded complex constructed here supplies the Stokes and power layers of a discrete field theory, but full multi-Dirac integrability is a further geometric property [34]. Finally, the product-Gauss uniqueness theorem is deliberately local and diagonal. The mixed sectors in Equation (37) show explicitly why nonlocal space–time Petrov–Galerkin pairings remain possible.

The visibility theorem is finite-dimensional and exact, but conditional on a fixed regular reconstruction. It classifies every volume two-form once the Dirac relation, constitutive law, boundary causality, and interior solve are fixed; it does not assert that these data are gauge independent. Proposition 4.16 identifies its precise first variation, while Theorem 4.14 proves exact raw realization for a gauge already encoded as $T_h^* S_h T_h$. The latter is not a derivation of S_h from independent volume dynamics. The missing interior mechanism is supplied in one direction by Theorem 8.8: the polynomial stage block is specified before the cap gauge and its Schur elimination produces that gauge. A second direction is now exact at the algebraic level: Theorem 9.10 proves that every rank-$2q$ cap deformation is the Schur reduction of exactly $2q$ auxiliary alternating modes. Because that generator is constructed from the target deformation, it is not yet a selector. The positive Stieltjes realization in Theorem 9.24 is more canonical

and produces the Gauss sequence, but Equation (168) proves that it cannot generate the full Taylor-corrected hierarchy. A fully coupled spatial–temporal cell in which independently specified mixed bubble modes select Θ, rather than encode it, therefore remains open in this sharper form. Extending Equation (42) to singular descriptor cells requires composition of linear Dirac relations. Extending it over a mesh with nontrivial topology requires separating exact, closed, and cohomological modes rather than labelling all transverse directions inactive.

The mixed transmission-line deformation proves structural nonseparability; Table 17 additionally demonstrates a concrete, but problem-specific, numerical gain. Its generic loss of symplecticity is quantified by Equation (141). A general selector balancing dispersion, causality, sparsity, and conditioning remains open. The global convergence table varies h at fixed spatial degree; it is not a uniform-in-N field convergence theorem. Nor is the scalar cap power identity by itself a local multisymplectic form formula on variations; deriving such a law requires an additional tangent cell analysis. Finally, the commutator test excludes the scalar functional calculus only after the stated spatial chart and boundary causality are fixed. It does not exclude matrix-valued coefficients, splittings with additional operators, or augmented problem-dependent formulations.

12 Conclusion

The discrete fundamental theorem of calculus and spatial integration by parts are boundary components of one affine Stokes problem on the discrete cylinder. The genuine total differential and the stacked directional differential encode this statement in different linear maps. For the former, a degree-respecting symmetric double has gauge $U_q^* \otimes V_q^*$; for the latter, the intrinsic active freedom is $\Lambda^2(E_{\mathrm{st}}/\ker D_{\mathrm{st}})^*$. The canonical filtration Equation (35) then identifies pure directions and the mixed quotient without pretending that a chosen direct-sum splitting is intrinsic.

A fixed regular interior solve transports this volume geometry to the caps by the exact map $\omega \mapsto L_h^*\omega$. Equation (42) determines all visible and invisible volume gauges and proves when every temporal gauge possesses a volume extension; Equation (46) isolates its mixed part canonically. A trace-generated volume form can be inserted in the raw weak cell, and Theorem 4.14 proves exact recovery of the prescribed cap gauge. This result is an exact realization theorem, not a hidden selection principle.

Within the finite-dimensional, nonsingular, quadratic setting, Theorem 5.3 identifies alternating gauges bijectively with the open subset of $O(Q)$ having no eigenvalue $+1$. These are constitutively shifted coordinates built from the classical Cayley transform; the group identity is excluded and the zero gauge represents midpoint. The exact flow, Padé maps, and mixed deformations are points in this cell; the symplectic centralizer is exactly the preimage of $O(Q) \cap Sp(\Omega)$. More importantly, Theorem 8.8 reverses the direction of construction: starting from the geometric Gauss pairing on $\mathcal{P}_s \to \mathcal{P}_{s-1}$, it eliminates sn active stage variables and derives both the diagonal Padé map and its constitutively adapted cap gauge. Thus Gauss–s is no longer merely encoded in a minimal two-point cell.

In space, Theorem 9.2 proves that the raw rectangular Stokes identity and an invertible effort/input reconstruction force the complete square open Dirac representation, including feedthrough. Applied to the 2012 transmission line, it combines with the active temporal cell and the canonical mixed lift $\mathcal{T}_N = \widetilde{\mathcal{T}}_N \pi_{tx}$. Boundary power, dissipation, and interface cancellation remain exact. At the unrestricted algebraic level, every desired rank-$2q$ change of cap gauge can be generated by a minimal $2q$-dimensional alternating auxiliary block. This proves finite-dimensional interior generability, but the block still encodes the target gauge and therefore does not constitute a field selector. For every s, the perturbation $h^{2s-1}\Theta$ is self-adjoint, energy preserving, and has exactly order $2s$ unless $\Theta = \Theta_s^\star = c_s \Omega_N A_N^{2s}$. This

unique fixed coefficient within the one-term ansatz raises the order to exactly $2s+2$; at $s=1$ it gives exactly Gauss–2. It is central, but this is not a high-order obstruction: a second arbitrary alternating term retains order $2s+2$ and, when noncommuting, separates the map from the scalar functional calculus of the semidiscrete line. The central choice $\Xi_s^\star = d_s \Omega_N A_N^{2s+2}$ raises the order by two more, and successive Taylor coefficients generate the explicit rational hierarchy Theorem 9.18, with global order at least $2s+2r$ and exactness conditional on $\alpha_{s,r} \neq 0$, converging locally to the exact-flow gauge. Its reduced form $\Pi_{s,r}(-z)^{-1}\Pi_{s,r}(z)$ separates removable $+1$ chart resonances from true propagator poles: there are no reduced poles on the imaginary axis, but the family is not generally A-stable. The Stieltjes representation of the exact coordinate proves the Padé status of C_s and gives each Gauss coordinate the sequence Equation (1), with positivity referring to the measure, residues, resolvent, and Jacobi coefficients rather than to the complete augmented port-Hamiltonian energy matrix. The corrected $(2,1)$ member is negative in the associated Stieltjes variable for $u > 60$, proving that this passive mechanism does not extend to the whole Taylor-corrected hierarchy. Strict non-annulation of all remaining coefficients is left as a precisely stated conjecture. The computations independently verify rectangular reconstruction, trace-generated realization, active-stage generation, stable Cayley evaluation, the coefficient d_s, the one-term dichotomy, the noncommuting high-order branch—including its $s=3$ multiprecision asymptotics—the reduced-pole distinction, exact coefficient-sign evidence through $s, j \leq 12$, and dispersion design. All numerical blocks are guarded by executable assertions. The remaining fundamental step is no longer algebraic existence of an interior generator; it is a fully coupled field cell whose mixed deformation is selected by spatial–temporal interior modes specified independently of the target gauge, together with a uniform space–time convergence and local multisymplectic analysis.

Acknowledgment

The author thanks the port-Hamiltonian and geometric-integration communities whose independent spatial, temporal, and field-theoretic constructions motivate the unified algebraic viewpoint developed here.

References

[1] S. Basu and N. K. Bose, Matrix Stieltjes series and network models, *SIAM Journal on Mathematical Analysis* 14 (1983), 209–222. doi:10.1137/0514018.

[2] T. J. Bridges and S. Reich, Multi-symplectic integrators: numerical schemes for Hamiltonian PDEs that conserve symplecticity, *Physics Letters A* 284 (2001), 184–193. doi:10.1016/S0375-9601(01)00294-8.

[3] L. Brugnano, F. Iavernaro, and D. Trigiante, Hamiltonian boundary value methods (energy preserving discrete line integral methods), *Journal of Numerical Analysis, Industrial and Applied Mathematics* 5 (2010), 17–37. https://arxiv.org/abs/0910.3621.

[4] A. Brugnoli, R. Rashad, and S. Stramigioli, Dual field structure-preserving discretization of port-Hamiltonian systems using finite element exterior calculus, *Journal of Computational Physics* 471 (2022), 111601. doi:10.1016/j.jcp.2022.111601.

[5] E. Celledoni and E. H. Høiseth, Energy-preserving and passivity-consistent numerical discretization of port-Hamiltonian systems, arXiv:1706.08621, 2017. https://arxiv.org/abs/1706.08621.

[6] E. Celledoni and J. Jackaman, Discrete conservation laws for finite element discretisations of multisymplectic PDEs, *Journal of Computational Physics* 444 (2021), 110520. doi:10.1016/j.jcp.2021.110520.

[7] P. Cheng, A high-order multi-symplectic finite element method for nonlinear wave equations, *Physica D: Nonlinear Phenomena* 496 (2026), 135329. doi:10.1016/j.physd.2026.135329.

[8] G. J. Cooper, Stability of Runge–Kutta methods for trajectory problems, *IMA Journal of Numerical Analysis* 7 (1987), 1–13. doi:10.1093/imanum/7.1.1.

[9] J. L. Cieśliński, Locally exact modifications of discrete gradient schemes, arXiv:1304.6533, 2013. https://arxiv.org/abs/1304.6533.

[10] S. Eidnes, Order theory for discrete gradient methods, *BIT Numerical Mathematics* 62 (2022), 1207–1255. doi:10.1007/s10543-022-00909-z.

[11] Z. Ge and J. E. Marsden, Lie–Poisson Hamilton–Jacobi theory and Lie–Poisson integrators, *Physics Letters A* 133 (1988), 134–139. doi:10.1016/0375-9601(88)90773-6.

[12] J. Giesselmann, A. Karsai, and T. Tscherpel, Energy-consistent Petrov–Galerkin time discretization of port-Hamiltonian systems, *The SMAI Journal of Computational Mathematics* 11 (2025), 335–367. doi:10.5802/smai-jcm.127.

[13] O. Gonzalez, Time integration and discrete Hamiltonian systems, *Journal of Nonlinear Science* 6 (1996), 449–467. doi:10.1007/BF02440162.

[14] E. Hairer, C. Lubich, and G. Wanner, *Geometric Numerical Integration: Structure-Preserving Algorithms for Ordinary Differential Equations*, 2nd ed., Springer, Berlin, 2006. doi:10.1007/3-540-30666-8.

[15] E. Hairer, Energy-preserving variant of collocation methods, *Journal of Numerical Analysis, Industrial and Applied Mathematics* 5 (2010), 73–84. https://www.unige.ch/~hairer/preprints/crete.html.

[16] T. Kemmochi, Higher order discrete gradient method by the discontinuous Galerkin time-stepping method, arXiv:2308.02334, 2023. https://arxiv.org/abs/2308.02334.

[17] P. Kotyczka, B. Maschke, and L. Lefèvre, Weak form of Stokes–Dirac structures and geometric discretization of port-Hamiltonian systems, *Journal of Computational Physics* 361 (2018), 442–476. doi:10.1016/j.jcp.2018.02.006.

[18] P. Kotyczka and L. Lefèvre, Discrete-time port-Hamiltonian systems: a definition based on symplectic integration, *Systems & Control Letters* 133 (2019), 104530. doi:10.1016/j.sysconle.2019.104530.

[19] B. Kostant and P. W. Michor, The generalized Cayley map from an algebraic group to its Lie algebra, in *The Orbit Method in Geometry and Physics*, Progress in Mathematics 213, Birkhäuser, 2003, 259–296. https://arxiv.org/abs/math/0109066.

[20] R. Moulla, L. Lefèvre, and B. Maschke, Pseudo-spectral methods for the spatial symplectic reduction of open systems of conservation laws, *Journal of Computational Physics* 231 (2012), 1272–1292. doi:10.1016/j.jcp.2011.10.008.

[21] R. I. McLachlan, G. R. W. Quispel, and N. Robidoux, Geometric integration using discrete gradients, *Philosophical Transactions of the Royal Society A* 357 (1999), 1021–1045. doi:10.1098/rsta.1999.0363.

[22] R. I. McLachlan and A. Stern, Multisymplecticity of hybridizable discontinuous Galerkin methods, *Foundations of Computational Mathematics* 20 (2020), 35–69. doi:10.1007/s10208-019-09415-1.

[23] J. E. Marsden, G. W. Patrick, and S. Shkoller, Multisymplectic geometry, variational integrators, and nonlinear PDEs, *Communications in Mathematical Physics* 199 (1998), 351–395. doi:10.1007/s002200050505.

[24] B. Wembe, C. Offen, S. Maslovskaya, S. Ober-Blöbaum, and P. Singh, Commutator-free Cayley methods, *Journal of Computational and Applied Mathematics* 477 (2026), 117184. doi:10.1016/j.cam.2025.117184.

[25] Y. Miyatake and J. C. Butcher, A characterization of energy-preserving methods and the construction of parallel integrators for Hamiltonian systems, *SIAM Journal on Numerical Analysis* 54 (2016), 1993–2013. doi:10.1137/15M1020861.

[26] G. R. W. Quispel and D. I. McLaren, A new class of energy-preserving numerical integration methods, *Journal of Physics A: Mathematical and Theoretical* 41 (2008), 045206. doi:10.1088/1751-8113/41/4/045206.

[27] S. Reich, Multi-symplectic Runge–Kutta collocation methods for Hamiltonian wave equations, *Journal of Computational Physics* 157 (2000), 473–499. doi:10.1006/jcph.1999.6372.

[28] J. M. Sanz-Serna, Runge–Kutta schemes for Hamiltonian systems, *BIT Numerical Mathematics* 28 (1988), 877–883. doi:10.1007/BF01954907.

[29] M. Seslija, A. J. van der Schaft, and J. M. A. Scherpen, Discrete exterior geometry approach to structure-preserving discretization of distributed-parameter port-Hamiltonian systems, *Journal of Geometry and Physics* 62 (2012), 1509–1531. doi:10.1016/j.geomphys.2012.02.006.

[30] P. Schulze, Structure-preserving time discretization of port-Hamiltonian systems via discrete gradient pairs, arXiv:2311.00403, 2023. https://arxiv.org/abs/2311.00403.

[31] A. Stern and E. Zampa, Multisymplecticity in finite element exterior calculus, *Foundations of Computational Mathematics* 26 (2026), 1809–1858. doi:10.1007/s10208-025-09720-y.

[32] A. Stern and E. Zampa, Finite element exterior calculus for time-dependent Hamiltonian partial differential equations, arXiv:2601.00103, 2026. https://arxiv.org/abs/2601.00103.

[33] A. J. van der Schaft and B. M. Maschke, Hamiltonian formulation of distributed-parameter systems with boundary energy flow, *Journal of Geometry and Physics* 42 (2002), 166–194. doi:10.1016/S0393-0440(01)00083-3.

[34] J. Vankerschaver, H. Yoshimura, and M. Leok, The Hamilton–Pontryagin principle and multi-Dirac structures for classical field theories, *Journal of Mathematical Physics* 53 (2012), 072903. doi:10.1063/1.4731481.

[35] H. S. Wall, *Analytic Theory of Continued Fractions*, Chelsea, New York, 1948.

[36] P. Wynn, Upon the Padé table derived from a Stieltjes series, *SIAM Journal on Numerical Analysis* 5 (1968), 805–834. doi:10.1137/0705060.